

MODULI OF VARIETIES

IZ TECÚM

ABSTRACT. SLMath notes.

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Date: September 4, 2026.

¹Based on lectures by Dori Bejleri [2] .

²Based on lectures by Kenneth Ascher [1] .

³Based on lectures by Kristin DeVleming [3] .

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⁵Based on lectures by Dori Bejleri [2] .

⁶Based on lectures by Kenneth Ascher [1] .

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Part 1. Stacks and Moduli⁷

1. MODULI FUNCTORS AND STACKS

Throughout these notes k is an algebraically closed field of characteristic zero, every scheme is of finite type over k , and $\mathcal{C} := \text{Sch}/k$ is the category of such schemes [19], [27, §2]. A moduli problem is a rule assigning to a scheme T the families over T of the objects being classified, and this section records what that rule has to be in order to be a stack. We write

$h_X := \text{Hom}_k(-, X)$ for the functor of points of $X \in \mathcal{C}$, $\text{PSh}(\mathcal{C}) := \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$ for presheaves of sets, $\mu_r := \ker(\mathbb{G}_m \xrightarrow{z \mapsto z^r} \mathbb{G}_m)$ the roots of unity, $BG := [\text{Spec } k/G]$ for an affine algebraic group $G \subseteq \text{GL}_m$, $\Theta := [\mathbb{A}^1/\mathbb{G}_m]$ the scaling quotient of the affine line, $\mathcal{M}_{g,n}$ the stack of stable n -pointed genus g curves.

Definition 1.1 (Functor of points). A presheaf $F \in \text{PSh}(\mathcal{C})$ is *representable* when $F \cong h_X$ for some $X \in \mathcal{C}$, and X is then a *fine moduli space* for F [11, §1], [27, §2].

Theorem 1.2 (Yoneda). Let $F \in \text{PSh}(\mathcal{C})$ and $f: Y \rightarrow X$. Evaluation at the identity is a bijection compatible with restriction,

$$\begin{array}{ccc} \text{Nat}(h_X, F) & \xrightarrow[\cong]{\eta \mapsto \eta_X(\text{id}_X)} & F(X) \\ \downarrow - \circ h_f & & \downarrow F(f) \\ \text{Nat}(h_Y, F) & \xrightarrow[\cong]{\eta \mapsto \eta_Y(\text{id}_Y)} & F(Y) \end{array}$$

so $X \mapsto h_X$ is fully faithful.

Proof. Naturality forces a transformation to be determined by its value on the identity, and no hypothesis beyond F being a presheaf is used.

$$\eta_T(g) = F(g)(\eta_X(\text{id}_X)) \quad (g \in h_X(T)) \xrightarrow{\text{naturality at } g} \eta \mapsto \eta_X(\text{id}_X) \text{ injective,}$$

$$\xi \in F(X), \quad \eta_T^\xi(g) := F(g)(\xi) \xrightarrow{(\eta^\xi)_X(\text{id}_X) = \xi} \eta \mapsto \eta_X(\text{id}_X) \text{ surjective}$$

[11, §1], [27, §2]. The two displays are mutually inverse assignments, and both routes around the square evaluate at f . $\square$

Example 1.3 (Hilbert functor). For $\mathbb{P}^N$ and a numerical polynomial P , let F_P send T to the flat families $Z \subseteq \mathbb{P}^N \times T$ with fibrewise Hilbert polynomial P . Then

$$F_P \cong h_{\text{Hilb}^P(\mathbb{P}^N)} \xrightarrow{1.1, [21, \S 1]} \text{Hilb}^P(\mathbb{P}^N) \text{ is a fine moduli space}$$

[11, §1].

Example 1.4 (Automorphisms obstruct representability). Let X carry a nontrivial $\sigma \in \text{Aut}(X)$ of order r , let $T' \rightarrow T$ be an étale μ_r -cover, and let $\mathcal{Y} \rightarrow T$ be the constant family twisted by σ . Then

$$\mathcal{Y} \times_T T' \cong X \times T', \quad \mathcal{Y} \not\cong X \times T \xrightarrow{1.2, 1.1} \text{no } F \cong h_M \text{ classifies these families,}$$

since a representable functor is a sheaf whose sections are determined locally [11, §1].

Definition 1.5 (Prestack and stack). A *prestack* over $\mathcal{C}$ is a category $\mathcal{X}$ fibred in groupoids over $\mathcal{C}$. It is a *stack* when, for every étale cover $\{T_i \rightarrow T\}$,

$$\text{Isom}_{\mathcal{X}}(a, b) \text{ is a sheaf on } \mathcal{C}/T \quad \text{and} \quad \text{every descent datum on } \{T_i\} \text{ is effective}$$

[11, §2].

⁷Based on lectures by Dori Bejleri [2].

Definition 1.6 (Algebraic stack). A stack $\mathcal{X}$ is *algebraic* when its diagonal $\Delta_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X} \times_k \mathcal{X}$ is representable, quasi-compact and separated, and some scheme U admits a smooth surjection onto $\mathcal{X}$, called an *atlas*,

$$U \times_{\mathcal{X}} U \rightrightarrows U \xrightarrow{u} \mathcal{X}$$

[11, §3].

Definition 1.7 (Inertia). The *inertia stack* of $\mathcal{X}$, whose T -points are pairs (a, φ) with $\varphi \in \text{Aut}_T(a)$, is the self-fibre product of the diagonal,

$$\begin{array}{ccc} I_{\mathcal{X}} & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \Delta_{\mathcal{X}} \\ \mathcal{X} & \xrightarrow{\Delta_{\mathcal{X}}} & \mathcal{X} \times_k \mathcal{X} \end{array}$$

and $\mathcal{X}$ is an algebraic space exactly when $I_{\mathcal{X}} \rightarrow \mathcal{X}$ is an isomorphism [11, §3].

1.1. Quotient Stacks. Every algebraic stack is étale locally a quotient stack, so quotients are the model case rather than a special case [9], [11, §3].

Definition 1.8 (Quotient stack). For an affine algebraic group G acting on $U \in \mathcal{C}$, the *quotient stack* $[U/G]$ sends T to the groupoid of G -torsors $P \rightarrow T$ with an equivariant $P \rightarrow U$, and $U \rightarrow [U/G]$ is an atlas as in Definition 1.6,

$$G \times U \xrightarrow[\text{pr}]{\text{act}} U \longrightarrow [U/G]$$

[11, §3].

Example 1.9 (Classifying stacks and the degeneration stack). A map $\Theta \rightarrow \mathcal{X}$ is a $\mathbb{G}_m$ -equivariant family over $\mathbb{A}^1$, so by Definition 1.8,

$$I_{BG} \cong G \times BG, \quad \text{Hom}(\Theta, \mathcal{X}) = (\text{one-parameter degenerations in } \mathcal{X})$$

[11, §3], [8, §2].

Proposition 1.10 (Atlas for stable curves). *Let $2g - 2 + n > 0$ and let $N = N(g, n)$ be large. Tricanonical embedding identifies $\mathcal{M}_{g,n}$ with a quotient of a locally closed $H \subseteq \text{Hilb}^P(\mathbb{P}^N)$,*

$$\begin{array}{ccc} \mathcal{U} & \longrightarrow & H \\ \downarrow & & \downarrow \\ \mathcal{M}_{g,n} & \xrightarrow{\cong} & [H/\text{PGL}_{N+1}] \end{array}$$

so $\mathcal{M}_{g,n}$ is a smooth proper Deligne–Mumford stack.

Proof. The hypothesis $2g - 2 + n > 0$ makes the tricanonical bundle ample on every stable curve, so embedding by it rigidifies the moduli problem and the only remaining choice is a projective frame.

$$\omega_{\mathcal{C}}^{\otimes 3}(\sum p_i) \text{ very ample} \xrightarrow{1.3, [21, \S 2]} H \text{ parameterizes embedded stable curves,} \quad \mathcal{M}_{g,n} \cong [H/\text{PGL}_{N+1}]$$

[11, §3]. The stabilizer at a point is the automorphism group of the curve, which stability makes finite and reduced, whence Deligne–Mumford. $\square$

Example 1.11 (Boundary of the stack of stable curves). The locus where the fibre acquires a node is a normal crossings divisor whose strata are indexed by stable dual graphs, and Figure 1 realizes one stratum inside $\mathcal{M}_{1,2}$ [11, §3], [21, §2].

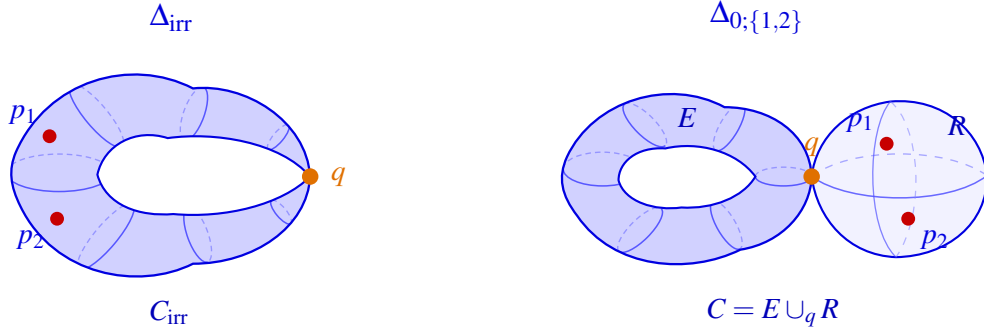


FIGURE 1. The divisor Δ_{irr} parametrizes irreducible nodal curves, whereas $\Delta_{0;\{1,2\}}$ parametrizes an elliptic component joined to a two-pointed rational tail.

Definition 1.12 (Root stack). Let $D \subseteq X$ be an effective Cartier divisor with section $s_D \in H^0(X, \mathcal{O}_X(D))$ and let $r \geq 1$. The r -th root stack of X along D is the fibre product

$$\begin{array}{ccc} X_{D,r} & \longrightarrow & \Theta \\ \downarrow & & \downarrow z \mapsto z^r \\ X & \xrightarrow{(\mathcal{O}_X(D), s_D)} & \Theta \end{array}$$

so $X_{D,r} \rightarrow X$ is an isomorphism off D and has stabilizer μ_r along D by Definition 1.7 [11, §3].

2. GOOD MODULI SPACES

A moduli stack retains automorphisms, a moduli space does not, and the correct comparison between the two is not a coarse space but a good moduli morphism. Throughout $\mathcal{Z}$ is an algebraic stack of finite type over k with affine stabilizers, Z is an algebraic space, and $\phi: \mathcal{Z} \rightarrow Z$. We also fix a discrete valuation ring R , that is a noetherian local domain whose maximal ideal $\mathfrak{m}_R = (\varpi)$ is principal, with uniformizer ϖ , fraction field $K = R[\varpi^{-1}]$, and residue field $\kappa = R/(\varpi)$ [19, §8].

Definition 2.1 (Good moduli morphism). A quasi-compact quasi-separated $\phi: \mathcal{Z} \rightarrow Z$ is a *good moduli morphism* when

$$(2.2) \quad \mathcal{O}_Z \xrightarrow{\cong} \phi_* \mathcal{O}_{\mathcal{Z}} \quad \text{and} \quad \phi_*: \text{QCoh}(\mathcal{Z}) \rightarrow \text{QCoh}(Z) \text{ is exact,}$$

and it is then initial among maps from $\mathcal{Z}$ to algebraic spaces [11, §6], [8, §2].

Example 2.3 (Geometric invariant theory). For a reductive G acting on $U = \text{Spec } A$, invariants are exact and finitely generated, so

$$[U/G] \longrightarrow \text{Spec } A^G \xrightarrow{(2.2)} \text{the affine quotient is a good moduli morphism}$$

[11, §6].

Theorem 2.4 (Closed points and S-equivalence). Let $\phi: \mathcal{Z} \rightarrow Z$ be a good moduli morphism with Z of finite type. Every fibre of ϕ contains a unique closed point, and

$$\phi(z_1) = \phi(z_2) \iff \overline{\{z_1\}} \cap \overline{\{z_2\}} \neq \emptyset \quad (z_1, z_2 \in |\mathcal{Z}|).$$

Proof. Exactness in (2.2) means closed substacks push forward to closed subspaces with the expected intersections, so images of point closures behave like the orbit closures of Example 2.3.

$$0 \rightarrow \mathcal{I}_{\mathcal{Z}'} \rightarrow \mathcal{O}_{\mathcal{Z}} \rightarrow \mathcal{O}_{\mathcal{Z}'} \rightarrow 0 \xrightarrow{(2.2)} \phi(\mathcal{Z}') \text{ closed,} \quad \phi(\mathcal{Z}_1 \cap \mathcal{Z}_2) = \phi(\mathcal{Z}_1) \cap \phi(\mathcal{Z}_2)$$

[11, §6], [8, §4]. Applying the second identity to the closures of z_1, z_2 gives the equivalence, and to two closed points of one fibre forces them to meet, hence to coincide. $\square$

Definition 2.5 (S-equivalence). Points $z_1, z_2 \in |\mathcal{Z}|$ are *S-equivalent* when their closures share a closed point, so by Theorem 2.4 the points of Z are the S-equivalence classes and the closed points of $\mathcal{Z}$ are the *polystable* objects [11, §6].

Theorem 2.6 (Luna étale slice for stacks). *Let $z \in |\mathcal{Z}|$ be a closed point with linearly reductive stabilizer G_z . Some affine G_z -scheme $\text{Spec} A$ fits in a cartesian étale neighbourhood*

$$\begin{array}{ccc} [\text{Spec} A / G_z] & \xrightarrow{\text{étale}} & \mathcal{Z} \\ \downarrow & & \downarrow \phi \\ \text{Spec} A^{G_z} & \xrightarrow{\text{étale}} & Z \end{array}$$

so every good moduli space is étale locally the quotient of Example 2.3.

Proof. [9], [11, §6]. $\square$

2.1. Existence Criteria. The two conditions below are valuative and are tested against the stacks Θ and $\mathcal{S}_R$ rather than against schemes. Θ -reductivity says a degeneration that stays inside the moduli problem has its limit inside the moduli problem, and S-completeness says two families over $\text{Spec} R$ with isomorphic generic fibres acquire a common closed degeneration, so together they replace reductivity of stabilizers and separatedness of the quotient [8, §3], [11, §6].

Definition 2.7 (The test stacks). With $\Theta = [\mathbb{A}^1 / \mathbb{G}_m]$ as in Example 1.9, set

$$(2.8) \quad \mathcal{S}_R := [\text{Spec}(R[s, t] / (st - \varpi)) / \mathbb{G}_m], \quad \deg(s) = 1, \quad \deg(t) = -1,$$

whose closed fibre is two branches meeting in a unique closed point, and write $\Theta^\circ, \mathcal{S}_R^\circ$ for the complements of the distinguished closed points [8, §3].

Definition 2.9 (Θ -reductivity and S-completeness). The stack $\mathcal{Z}$ is *Θ -reductive*, respectively *S-complete*, when every map from the punctured stack of (2.8) extends uniquely up to unique 2-isomorphism,

$$\begin{array}{ccccc} \Theta_R^\circ & \longrightarrow & \mathcal{Z} & \longleftarrow & \mathcal{S}_R^\circ \\ \downarrow & \nearrow \exists! & & \nwarrow \exists! & \downarrow \\ \Theta_R & & & & \mathcal{S}_R \end{array}$$

[8, §3].

Theorem 2.10 (Alper–Halpern–Leistner–Heinloth). *Let $\mathcal{Z}$ be of finite presentation over k with affine stabilizers and separated diagonal. Then*

$$\mathcal{Z} \text{ admits a separated good moduli space} \xLeftrightarrow[2.9, [8, \S 1]]{\text{}} \mathcal{Z} \text{ is } \Theta\text{-reductive and S-complete,}$$

and the good moduli space is proper exactly when $\mathcal{Z}$ also satisfies the existence part of the valuative criterion over $\text{Spec} R$.

Proof. [8, §1], [11, §6]. $\square$

Part 2. Algebraic Surfaces⁸

3. INTERSECTION THEORY AND POSITIVITY

Birational geometry asks for a distinguished model inside a birational class, and for surfaces the answer is computed entirely by intersection numbers. From here to the end of Part III, X is a smooth projective surface over k , and $D_1 \equiv D_2$ means $(D_1 - D_2) \cdot C = 0$ for every integral curve $C \subseteq X$ [19, §V.1], [22, §1]. We write

$$\begin{aligned} N^1(X)_{\mathbb{R}} &:= \text{NS}(X) \otimes_{\mathbb{Z}} \mathbb{R} \text{ for the real Néron–Severi group,} & N_1(X)_{\mathbb{R}} &:= Z_1(X)_{\mathbb{R}} / \equiv \text{ for real 1-cycles,} \\ \rho(X) &:= \dim_{\mathbb{R}} N^1(X)_{\mathbb{R}} \text{ the Picard number,} & \text{Nef}(X) &:= \{D \mid D \cdot C \geq 0 \text{ for every curve } C\}, \\ \overline{\text{NE}}(X) &:= \overline{\{\sum a_i [C_i] \mid a_i \geq 0\}} \text{ the Mori cone,} & K_X &\text{ the class of the canonical sheaf } \omega_X = \Omega_{X/k}^2, \\ \chi(D) &:= \chi(X, \mathcal{O}_X(D)), & p_a(C) &:= 1 - \chi(\mathcal{O}_C) \text{ the arithmetic genus,} & R(X, K_X) &:= \bigoplus_{m \geq 0} H^0(X, mK_X), \end{aligned}$$

the last being the canonical ring, the central birational invariant.

Theorem 3.1 (Intersection pairing). *There is a unique symmetric bilinear pairing $\text{Div}(X) \times \text{Div}(X) \rightarrow \mathbb{Z}$ with $C \cdot D = \#(C \cap D)$ for smooth curves meeting transversally, and it factors through $\equiv$ to a perfect pairing*

$$N^1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}.$$

Proof. [19, §V.1]. □

Theorem 3.2 (Riemann–Roch and Serre duality). *For every $D \in \text{Div}(X)$,*

$$(3.3) \quad \chi(D) = \chi(\mathcal{O}_X) + \frac{1}{2} D \cdot (D - K_X), \quad h^2(X, \mathcal{O}_X(D)) = h^0(X, \mathcal{O}_X(K_X - D)).$$

Proof. [19, §V.1]. □

Theorem 3.4 (Adjunction). *For an integral curve $C \subseteq X$ the dualizing sheaf is $\omega_C = (\omega_X \otimes \mathcal{O}_X(C))|_C$, so*

$$2p_a(C) - 2 = C \cdot (C + K_X).$$

Proof. [19, §V.1], [22, §1]. □

Corollary 3.5 (The two negative curves). *For $C \cong \mathbb{P}^1$,*

$$C^2 = -1 \xrightarrow{3.4} K_X \cdot C = -1, \quad C^2 = -2 \xrightarrow{3.4} K_X \cdot C = 0.$$

The first class is removed by the program of §7, the second survives on a minimal model and is contracted by Theorem ??.

Proof. Both cases are adjunction for a rational curve, where $p_a(C) = 0$.

$$-2 = 2p_a(C) - 2 \xleftrightarrow{3.4} C^2 + K_X \cdot C = -2.$$

Substituting $C^2 = -1$ and $C^2 = -2$ gives the two K_X -degrees. □

Theorem 3.6 (Hodge index). *Let H be ample. The intersection form on $N^1(X)_{\mathbb{R}}$ has signature $(1, \rho(X) - 1)$, so*

$$(D \cdot H)^2 \geq D^2 H^2, \quad D \cdot H = 0, \quad D \neq 0 \xrightarrow{3.1} D^2 < 0,$$

with equality on the left exactly when D and H are numerically proportional.

Proof. [19, §V.1]. □

Theorem 3.7 (Nakai–Moishezon and Kleiman). *For $D \in N^1(X)_{\mathbb{R}}$,*

$$D \text{ ample} \iff D^2 > 0 \text{ and } D \cdot C > 0 \quad (\text{all integral } C \subseteq X) \xleftrightarrow{[22, \S 1]} D \in \text{int Nef}(X).$$

Proof. [19, §V.1], [22, §1]. □

⁸Based on lectures by Kenneth Ascher [1].

3.1. Zariski Decomposition. A pseudo-effective class fails to be nef only along finitely many curves of negative self-intersection, and Hodge index makes the correction unique.

Theorem 3.8 (Zariski decomposition). *Every pseudo-effective $\mathbb{Q}$ -divisor D on X has a unique decomposition into a nef part and a negative-definite effective part,*

$$D = P + N, \quad N = \sum_i a_i N_i, \quad (N_i \cdot N_j) \text{ negative definite}, \quad P \cdot N_i = 0 \quad (\text{all } i),$$

and then $\text{vol}(D) = P^2$, the fixed part of $|mD|$ being mN for sufficiently divisible m ; see Figure 2.

Proof. Every curve met negatively by D must sit in N , and Hodge index controls how many such curves can coexist.

$$D \cdot N_i < 0 \xrightarrow{3.6} (N_i \cdot N_j) \text{ negative definite} \xrightarrow{3.1} P \cdot N_i = 0 \text{ has a unique solution,}$$

$$N^1(X)_{\mathbb{R}} \text{ finite dimensional} \xrightarrow{[22, \S 2]} \text{the enlargement terminates} \xrightarrow{3.7} \text{vol}(D) = P^2.$$

The support of N is assembled in finitely many enlargements, and orthogonality of P against it identifies the volume with P^2 . $\square$

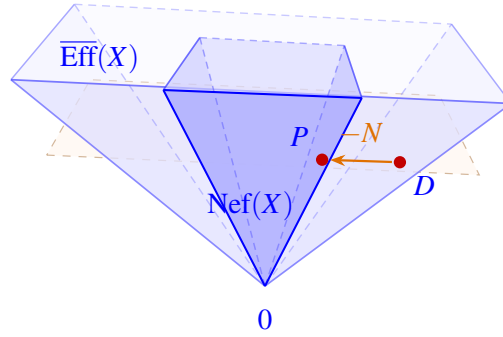


FIGURE 2. Zariski decomposition writes $D = P + N$, with P on the nef face determined by the negative part N .

4. BLOW-UPS AND RULED SURFACES

Every birational morphism between smooth surfaces factors into blow-ups of points, so a single local construction and its numerical shadow control the whole birational calculus, and ruled surfaces supply the standard stock of extremal contractions. Throughout, $\text{Bl}_p X$ is the blow-up of the point $p \in X$, E its exceptional curve, $\tilde{C}$ the strict transform of a curve C , and on the Hirzebruch surface $\mathbb{F}_n$ the classes S and F are the section of least self-intersection and a fibre of the ruling [19, §V.2–V.3].

Construction 4.1 (Blow-up of a point). For a smooth point $p \in X$ let $\pi: \tilde{X} = \text{Bl}_p X \rightarrow X$ be the *blow-up* with *exceptional curve* $E = \pi^{-1}(p) \cong \mathbb{P}^1$, which parameterizes tangent directions at p . Then

$$(4.2) \quad E^2 = -1, \quad \pi^* D \cdot E = 0, \quad \pi^* D \cdot \pi^* D' = D \cdot D', \quad K_{\tilde{X}} = \pi^* K_X + E,$$

and for C of multiplicity m at p , with *strict transform* $\tilde{C} = \pi^* C - mE$,

$$\tilde{C}^2 = C^2 - m^2, \quad K_{\tilde{X}} \cdot \tilde{C} = K_X \cdot C + m, \quad K_{\tilde{X}}^2 = K_X^2 - 1, \quad \rho(\tilde{X}) = \rho(X) + 1,$$

so $N^1(\tilde{X}) = \pi^* N^1(X) \oplus \mathbb{Z}E$ [19, §V.3]; see Figure 3.

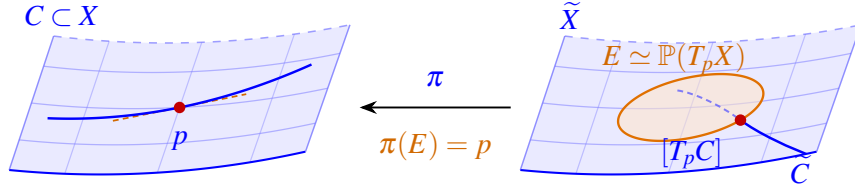


FIGURE 3. The blow-up replaces p by $E \simeq \mathbb{P}(T_p X)$, and $\tilde{C}$ meets it transversely at $[T_p C]$.

Example 4.3 (Plane curves). For $C \subseteq \mathbb{P}^2$ of degree d with multiplicity m at p , and $H = \pi^* \mathcal{O}_{\mathbb{P}^2}(1)$,

$$\tilde{C} \equiv dH - mE, \quad \tilde{C}^2 = d^2 - m^2, \quad K_{\text{Bl}_p \mathbb{P}^2} = -3H + E \xrightarrow{(4.2)} K_{\text{Bl}_p \mathbb{P}^2}^2 = 8.$$

Theorem 4.4 (Castelnuovo contraction). *Let $E \subseteq X$ satisfy $E \simeq \mathbb{P}^1$ and $E^2 = -1$. Then E is contracted to a smooth point of a smooth projective surface by a birational morphism π , and conversely,*

$$X \cong \text{Bl}_p Y, \quad E = \pi^{-1}(p).$$

Proof. Adjunction pins the normal bundle of E to the local model of a blow-up, and a linear system constant on E performs the contraction.

$$E \simeq \mathbb{P}^1, \quad E^2 = -1 \xrightarrow{3.5} K_X \cdot E = -1, \quad \mathcal{O}_E(E) \simeq \mathcal{O}_{\mathbb{P}^1}(-1) \xrightarrow{(4.2)} \text{the blow-up local model,}$$

$$\text{a suitable } |H| \text{ is constant on } E \text{ and injective off } E \xrightarrow{[19, \S V.5]} X \cong \text{Bl}_p Y.$$

The image point is smooth because the normal bundle matches the local model, and the converse is (4.2). $\square$

Theorem 4.5 (Factorization). *Every birational morphism of smooth projective surfaces is a composite of blow-ups of points, and every birational map ϕ admits a common resolution by such composites,*

$$\begin{array}{ccc} & Z & \\ p \swarrow & & \searrow q \\ X & \xrightarrow{\phi} & Y \end{array}$$

Proof. Both claims reduce to resolving the finitely many base points of the linear system defining ϕ .

resolve the base points of $\phi \xrightarrow{[19, \S V.5]} Z \rightarrow Y$ a morphism, repeat for $\phi^{-1} \xrightarrow{4.4} p, q$ composites of blow-ups.

Each resolution step is the blow-up of a point, so both projections factor as claimed. $\square$

Theorem 4.6 (Noether–Castelnuovo). *The birational automorphism group of $\mathbb{P}^2$ is generated by PGL_3 and the standard quadratic Cremona transformation*

$$\sigma: \mathbb{P}^2 \dashrightarrow \mathbb{P}^2, \quad [x:y:z] \mapsto [yz:xz:xy],$$

resolved by blowing up $p_1 = [1:0:0]$, $p_2 = [0:1:0]$, $p_3 = [0:0:1]$; see Figure 4.

Proof. A homaloidal system of degree d with base multiplicities m_i satisfies the Noether equalities, and the three largest multiplicities are the centres for a degree-lowering composition.

$$\sum_i m_i^2 = d^2 - 1, \quad \sum_i m_i = 3d - 3, \quad d > 1 \xrightarrow{(4.2), 3.4} m_1 + m_2 + m_3 > d,$$

$$\text{compose with } \sigma \text{ centred at those three points} \xrightarrow{4.5} \text{the degree drops} \xrightarrow{\text{induction}} d = 1.$$

Each quadratic composition strictly lowers the degree, so the induction terminates in the linear group [19, §V.5], [22, §1]. $\square$

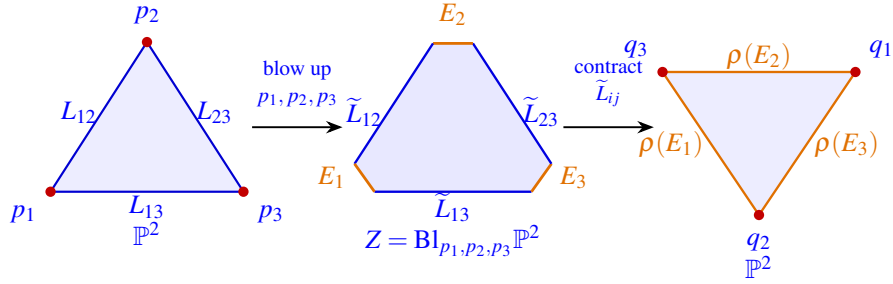


FIGURE 4. The quadratic Cremona map is resolved by replacing the three base points with the exceptional curves E_i and then contracting the strict transforms $\tilde{L}_{ij}$ of the coordinate lines.

Definition 4.7 (Hirzebruch surfaces). For $n \geq 0$ the *Hirzebruch surface* is $\mathbb{F}_n := \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(n)) \rightarrow \mathbb{P}^1$, with S the section of least self-intersection and F a fibre, so

$$(4.8) \quad \text{Pic}(\mathbb{F}_n) = \mathbb{Z}S \oplus \mathbb{Z}F, \quad S^2 = -n, \quad S \cdot F = 1, \quad F^2 = 0, \quad K_{\mathbb{F}_n} = -2S - (n+2)F,$$

and for $D = aS + bF$,

$$D \cdot F = a, \quad D \cdot S = b - an \xrightarrow{3.7} \overline{\text{NE}}(\mathbb{F}_n) = \mathbb{R}_{\geq 0}[S] + \mathbb{R}_{\geq 0}[F], \quad \text{Nef}(\mathbb{F}_n) = \mathbb{R}_{\geq 0}[F] + \mathbb{R}_{\geq 0}[S + nF]$$

[19, §V.2].

Construction 4.9 (Elementary transformation). For $p \in \mathbb{F}_n$ on a fibre F_p , the *elementary transformation* blows up p and contracts the strict transform $\tilde{F}_p$, a (-1) -curve by (4.2). The result is again ruled, with

$$p \in S \xrightarrow{4.4} \mathbb{F}_{n+1}, \quad p \notin S \xrightarrow{4.4} \mathbb{F}_{n-1},$$

see Figure 5. This is why minimal models are not unique when K_X is not pseudo-effective.

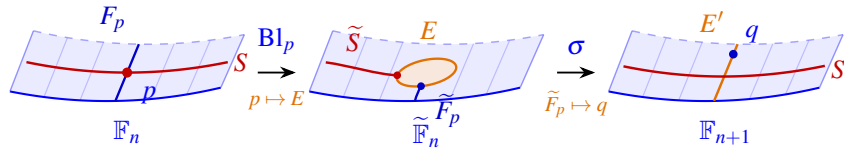


FIGURE 5. Blowing up $p = S \cap F_p$ separates $\tilde{S}$ from $\tilde{F}_p$, and contracting $\tilde{F}_p$ turns E into a fiber of $\mathbb{F}_{n+1}$.

Example 4.10 (The crepant contraction on $\mathbb{F}_2$). On $\mathbb{F}_2$ the divisor $H = S + 2F$ is nef and base-point free with

$$H \cdot S = 0, \quad H^2 = 2, \quad K_{\mathbb{F}_2} \cdot S = 0 \xrightarrow{3.5, (4.8)} |H| \text{ contracts } S \text{ crepantly,}$$

the image being the quadric cone $Q \subseteq \mathbb{P}^3$ with an A_1 singularity. This is a step from a minimal model to a canonical model, not a K -negative step of the program in §7.

4.1. Del Pezzo Surfaces. Del Pezzo surfaces are the two-dimensional Fano varieties, and they are the running examples for stability in Parts V and VI.

Definition 4.11 (Del Pezzo surface). A smooth projective surface X is *del Pezzo* when $-K_X$ is ample, and its *degree* is $d := K_X^2 \in \{1, \dots, 9\}$ [22, §1].

Theorem 4.12 (Classification). *Every del Pezzo surface is $\mathbb{P}^1 \times \mathbb{P}^1$ or $\text{Bl}_{p_1, \dots, p_r} \mathbb{P}^2$ with $0 \leq r \leq 8$ points in general position, and then*

$$-K_X = 3H - \sum_{i=1}^r E_i \xrightarrow{(4.2)} d = K_X^2 = 9 - r,$$

the Mori cone being generated by the finitely many (-1) -curves.

Proof. [22, §1], [19, §V.4]. □

Proposition 4.13 (Anticanonical models). *For $d \geq 3$ the class $-K_X$ is very ample and embeds X as a surface of degree d ,*

$$X \xhookrightarrow{|-K_X|} \mathbb{P}^d = \mathbb{P}(H^0(X, -K_X)^\vee) \quad h^0(X, -K_X) = d + 1.$$

Proof. Riemann–Roch computes the Euler characteristic and Kodaira vanishing removes the higher cohomology, so only very ampleness needs a reference.

$$\chi(-K_X) = \chi(\mathcal{O}_X) + \frac{1}{2}(-K_X) \cdot (-2K_X) \xleftrightarrow{(3.3)} \chi(-K_X) = 1 + d, \quad H^i(X, -K_X) = 0 \quad (i > 0).$$

The dimension count gives the embedding of the stated degree, and very ampleness for $d \geq 3$ is [19, §V.4], [22, §1]. □

Example 4.14 (Degrees seven and three). For $r = 2$ the three (-1) -curve classes are E_1 , E_2 and $H - E_1 - E_2$, and $K_X^2 = 7$. For $r = 6$, Proposition 4.13 realizes X as a smooth cubic $X \subseteq \mathbb{P}^3$ with

$$K_X = \mathcal{O}_X(-1) \xrightarrow{3.4} (-K_X)^2 = 3,$$

whose 27 lines are the six E_i , the fifteen classes $H - E_i - E_j$ and the six classes $2H - \sum_{l \neq i} E_l$. Cubic surfaces reappear in §6 as the model case where stability and invariant theory agree [19, §V.4], [15, §5].

Part 3. K-Stability⁹

5. TEST CONFIGURATIONS AND VALUATIVE CRITERIA

Fixing dimension and volume bounds nothing for Fano varieties, so the stable-pair moduli problem of §9 has no Fano analogue until a stability condition is imposed. Throughout X is a $\mathbb{Q}$ -Fano variety, meaning a normal projective variety with klt singularities as in Definition 8.4 such that $-K_X$ is $\mathbb{Q}$ -Cartier and ample, and $n = \dim X$. We write

$$\begin{aligned} V &:= (-K_X)^n \text{ the anticanonical volume,} & \text{Val}_{x,X} &\text{ the real valuations of } k(X) \text{ centred at } x, \\ \mathfrak{a}_k(v) &:= \{f \in \mathcal{O}_{X,x} \mid v(f) \geq k\} \text{ the valuation ideals in the local ring } \mathcal{O}_{X,x}, \\ \overline{\mathcal{X}} &\text{ the compactification of } \mathcal{X} \text{ over } \mathbb{P}^1, & \mathcal{M}_{n,V}^{K\text{-ss}}, M_{n,V}^K &\text{ the K-moduli stack and space.} \end{aligned}$$

Theorem 5.1 (Yau–Tian–Donaldson). *Let X be a smooth Fano manifold. Then*

$$\exists \omega \text{ Kähler with } \text{Ric}(\omega) = \omega \iff X \text{ K-polystable.}$$

Proof. [26, §1], [15, §1]. □

Example 5.2 (Unboundedness of klt Fano surfaces). Let (a, b, c) solve the Markov equation and mutate it,

$$(5.3) \quad a^2 + b^2 + c^2 = 3abc, \quad (a, b, c) \mapsto (a, b, 3ab - c),$$

starting from $(1, 1, 1)$. For the well-formed weighted projective plane $X = \mathbb{P}(a^2, b^2, c^2)$,

$$K_X \sim \mathcal{O}_X(-a^2 - b^2 - c^2) \xrightarrow{(5.3)} (-K_X)^2 = \frac{(a^2 + b^2 + c^2)^2}{a^2 b^2 c^2} = 9,$$

and each such surface is klt with a $\mathbb{Q}$ -Gorenstein smoothing to $\mathbb{P}^2$. So klt Fano surfaces of volume 9 are neither bounded nor separated; see Figure 6. Nonreductive automorphism groups and nonunique birational models are the two further failures that stability repairs [15, §1].

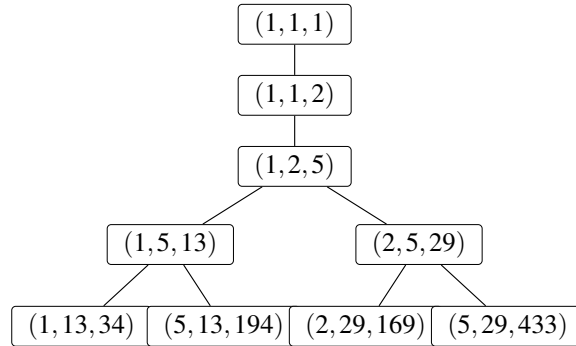


FIGURE 6. The first levels of the Markov mutation tree modulo coordinate permutations, rooted at $(1, 1, 1)$.

Definition 5.4 (Special test configuration). A *special test configuration* of X is a flat projective $\pi: \mathcal{X} \rightarrow \mathbb{A}^1$ with a $\mathbb{G}_m$ -action lifting the standard action, such that

$$\mathcal{X}|_{\mathbb{A}^1 \setminus \{0\}} \cong X \times (\mathbb{A}^1 \setminus \{0\}) \text{ } \mathbb{G}_m\text{-equivariantly,} \quad X_0 \text{ } \mathbb{Q}\text{-Fano,} \quad -K_{\mathcal{X}/\mathbb{A}^1} \text{ relatively ample,}$$

recorded after clearing denominators by the linearization $\mathcal{L} \sim -mK_{\mathcal{X}/\mathbb{A}^1}$; see Figure 7. Arbitrary test configurations reduce to special ones by base change, normalization and a relative minimal model program as in Theorem 8.15, without increasing the invariant of Definition 5.5 [25].

⁹Based on lectures by Kristin DeVleming [3].

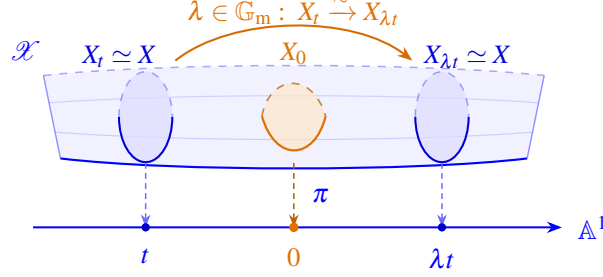


FIGURE 7. A special test configuration $\pi : \mathcal{X} \rightarrow \mathbb{A}^1$ is $\mathbb{G}_m$ -equivariantly trivial away from 0, while the distinguished fiber X_0 records the degeneration.

Definition 5.5 (Futaki invariant). For a polarized test configuration $(\mathcal{X}, \mathcal{L})$ with $d_k = h^0(X, L^k) = a_0 k^n + a_1 k^{n-1} + O(k^{n-2})$ and total $\mathbb{G}_m$ -weight $w_k = b_0 k^{n+1} + b_1 k^n + O(k^{n-1})$ on the central fibre, the *Futaki invariant* is $\text{Fut}(\mathcal{X}, \mathcal{L}) := (a_1 b_0 - a_0 b_1) / a_0^2$, and for a special test configuration

$$(5.6) \quad \text{Fut}(\mathcal{X}, \mathcal{L}) = -\frac{1}{2V(n+1)} (-K_{\mathcal{X}/\mathbb{P}^1})^{n+1}.$$

Definition 5.7 (K-stability). The $\mathbb{Q}$ -Fano variety X is *K-semistable*, *K-stable*, respectively *K-polystable*, according as

$$\text{Fut} \geq 0 \text{ always,} \quad \text{Fut} = 0 \implies \text{trivial configuration,} \quad \text{Fut} = 0 \implies \mathcal{X} \cong X \times \mathbb{A}^1,$$

over all special test configurations of Definition 5.4, the last case allowing a nontrivial action on the factor X [15, §2], [25].

Example 5.8 (The projective line). Every product test configuration of $\mathbb{P}^1$ compactifies to a Hirzebruch surface $\mathbb{F}_m$ as in Definition 4.7, and

$$K_{\mathbb{F}_m}^2 = 8 \xrightarrow{(5.6), (4.8)} \text{Fut} = 0, \quad \text{Aut}(\mathbb{P}^1) = \text{PGL}_2 \text{ positive dimensional} \xrightarrow{5.7} \mathbb{P}^1 \text{ K-polystable, not K-stable.}$$

Theorem 5.9 (First consequences).

$$X \text{ K-semistable} \xrightarrow{[25]} X \text{ klt}, \quad X \text{ K-stable} \xrightarrow{[7]} \text{Aut}(X) \text{ finite}, \quad X \text{ K-polystable} \xrightarrow{[7]} \text{Aut}(X) \text{ reductive.}$$

Proof. [25], [7], [15, §2]. □

5.1. The Delta Invariant. Definition 5.7 quantifies over infinitely many degenerations, and thresholds and valuations replace them by a single infimum over divisors over X , which is what makes stability computable.

Definition 5.10 (Tian's alpha invariant). For an algebraic group G acting on X , and with lct as in Definition 8.7, the *alpha invariant* and its equivariant refinement are

$$\alpha(X) := \inf_{0 \leq D \sim_{\mathbb{Q}} -K_X} \text{lct}(X, D), \quad \alpha_G(X) := \inf_{\substack{0 \leq D \sim_{\mathbb{Q}} -K_X \\ D \text{ G-invariant}}} \text{lct}(X, D)$$

[26, §1].

Theorem 5.11 (Tian's criterion).

$$\alpha(X) > \frac{n}{n+1} \xrightarrow{[26, \S 1]} X \text{ K-stable}, \quad \alpha(X) \geq \frac{n}{n+1} \xrightarrow{[26, \S 1]} X \text{ K-semistable},$$

and for G reductive with $\alpha_G(X) > n/(n+1)$, X is *K-polystable*.

Proof. [26, §1], [15, §3]. □

Remark 5.12 (The criterion is not necessary). $\alpha(\mathbb{P}^n) = 1/(n+1)$ while $\mathbb{P}^n$ is K -polystable, and no nonzero divisor is PGL_{n+1} -invariant, so $\alpha_{\text{PGL}_{n+1}}(\mathbb{P}^n) = \infty$ and the equivariant form of Theorem 5.11 applies [15, §3].

Example 5.13 (Degree-one del Pezzo surfaces). Let $X = \text{Bl}_{p_1, \dots, p_8} \mathbb{P}^2$ with the points in general position, so $(-K_X)^2 = 1$ by Theorem 4.12. For $D \sim_{\mathbb{Q}} -K_X$ with $\text{Supp} D$ not in the anticanonical pencil, choose $C \in |-K_X|$ through $x \in \text{Supp} D$.

$$D \cdot C = (-K_X)^2 = 1 \xrightarrow{8.5} \text{mult}_x D \leq 1 \xrightarrow{8.4} (X, D) \text{ log canonical,}$$

and the reduced members of the pencil are smooth, nodal or cuspidal with thresholds $1, 1, 5/6$ by Example 8.9, so

$$\alpha(X) \geq \frac{5}{6} > \frac{2}{3} \xrightarrow{5.11} X \text{ K-stable.}$$

Definition 5.14 (Volume of a divisor). For a $\mathbb{Q}$ -Cartier divisor D on a normal projective n -fold, the *volume* is

$$\text{vol}(D) := \limsup_{m \rightarrow \infty} \frac{h^0(X, \mathcal{O}_X(\lfloor mD \rfloor))}{m^n/n!},$$

and D is *big* when $\text{vol}(D) > 0$, with $\text{vol}(D) = D^n$ for D nef and $\text{vol}(D) = P^2$ on a surface, for the Zariski decomposition of Theorem 3.8.

Definition 5.15 (The invariants A , S , β and δ). For a prime divisor E over X appearing on $\mu: Y \rightarrow X$, with $A_X(E) = A_{X,0}(E)$ as in Definition 8.2, the *expected vanishing order* and *pseudo-effective threshold* are

$$(5.16) \quad S_X(E) := \frac{1}{V} \int_0^{\tau(E)} \text{vol}(\mu^*(-K_X) - tE) dt, \quad \tau(E) := \sup \{t \geq 0 \mid \mu^*(-K_X) - tE \text{ big}\},$$

and the *stability threshold* of X is

$$\beta_X(E) := A_X(E) - S_X(E), \quad \delta_X(E) := \frac{A_X(E)}{S_X(E)}, \quad \delta(X) := \inf_E \delta_X(E),$$

the integrand vanishing beyond $\tau(E)$ by Definition 5.14, so the integral is finite [17], [14].

Theorem 5.17 (Fujita–Li valuative criterion).

$$X \text{ K-semistable} \xLeftrightarrow{[17], [24]} \beta_X(E) \geq 0 \quad (\text{all prime divisors } E \text{ over } X) \xLeftrightarrow{5.15} \delta(X) \geq 1,$$

and the strict inequalities characterize K -stability.

Proof. [17], [24], [14]. □

Example 5.18 (The first Hirzebruch surface is unstable). For $X = \mathbb{F}_1$ with Σ the negative section and F a fibre, Definition 4.7 gives $-K_X = 2\Sigma + 3F$ and $V = 8$. With $D_t = -K_X - t\Sigma$,

$$D_t \cdot \Sigma = 1 + t, \quad D_t \cdot F = 2 - t \xrightarrow{3.7} D_t \text{ nef for } 0 \leq t \leq 2, \quad \tau(\Sigma) = 2,$$

$$S_X(\Sigma) = \frac{1}{8} \int_0^2 D_t^2 dt = \frac{1}{8} \int_0^2 (8 - 2t - t^2) dt = \frac{7}{6}, \quad A_X(\Sigma) = 1 \xrightarrow{(5.16), 5.17} \delta_X(\Sigma) = \frac{6}{7} < 1,$$

so $\mathbb{F}_1$ is K -unstable.

Theorem 5.19 (Openness). For a $\mathbb{Q}$ -Gorenstein flat family of $\mathbb{Q}$ -Fano varieties,

$$t \longmapsto \min\{1, \delta(X_t)\} \text{ is constructible and lower semicontinuous,}$$

so K -semistability is an open condition.

Proof. [14], [15, §4]. □

6. K-MODULI

The three failures of Example 5.2 are repaired by boundedness, openness and properness of the K-semistable locus, and the resulting stack has a good moduli space in the sense of §2. The local invariant that drives boundedness is a normalized volume at a singular point, computed in the local ring $R = \mathcal{O}_{X,x}$ through the valuation ideals $\mathfrak{a}_k(v)$ of §5.

Definition 6.1 (Normalized volume). For an n -dimensional klt singularity $x \in X$ and $v \in \text{Val}_{x,X}$, the volume, normalized volume and local volume are

$$\text{vol}(v) := \lim_{k \rightarrow \infty} \frac{\text{length}(R/\mathfrak{a}_k(v))}{k^n/n!}, \quad \widehat{\text{vol}}(v) := A_X(v)^n \text{vol}(v), \quad \widehat{\text{vol}}(x, X) := \inf_{v \in \text{Val}_{x,X}} \widehat{\text{vol}}(v),$$

the exponent n removing the scaling ambiguity because $A_X(cv) = cA_X(v)$ and $\text{vol}(cv) = c^{-n}\text{vol}(v)$ [24].

Proposition 6.2 (Basic values). For an n -dimensional klt singularity $x \in X$,

$$\widehat{\text{vol}}(x, X) \leq n^n \xLeftrightarrow[6.1] x \text{ smooth}, \quad x \in \mathbb{A}^n/G, \ G \subseteq \text{GL}_n \text{ free in codimension one} \xRightarrow[6.1]{} \widehat{\text{vol}}(x, X) = \frac{n^n}{|G|},$$

the first with equality exactly at a smooth point, and for $n = 2, 3$ a singular point satisfies $\widehat{\text{vol}}(x, X) \leq 2(n-1)^n$ with equality exactly at an ordinary double point.

Proof. [24], [15, §5]. □

Theorem 6.3 (Volume minimization at a cone). Let V be $\mathbb{Q}$ -Fano, let $X = C(V, -rK_V)$ be the affine cone with vertex o , and let $V_0 \cong V$ be the exceptional divisor of the blow-up of o . Then

$$V \text{ K-semistable} \xLeftrightarrow[24] \text{ord}_{V_0} \text{ minimizes } \widehat{\text{vol}} \text{ among valuations centred at } o;$$

see Figure 8.

Proof. [24], [25]. □

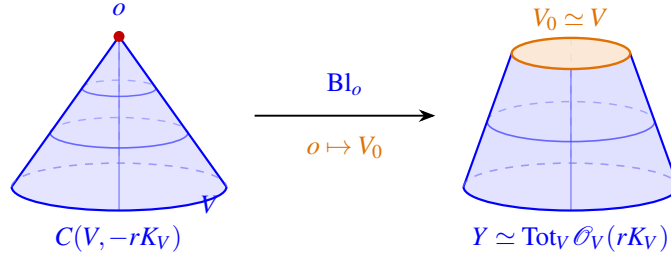


FIGURE 8. Blowing up the cone vertex extracts the exceptional zero section $V_0 \simeq V \subset Y$, whose valuation ord_{V_0} is the canonical cone valuation in normalized-volume criteria for K-semistability.

Theorem 6.4 (Local-to-global volume bound). For X K-semistable $\mathbb{Q}$ -Fano of dimension n and every $x \in X$,

$$(6.5) \quad (-K_X)^n \leq \left(1 + \frac{1}{n}\right)^n \widehat{\text{vol}}(x, X).$$

Proof. The sections of $-mK_X$ vanishing to valuation order mt estimate the volume of the twist, and K-semistability enters only through $\beta_X(v) \geq 0$.

$$0 \rightarrow \mathcal{O}_X(-mK_X)\mathfrak{a}_{mt} \rightarrow \mathcal{O}_X(-mK_X) \rightarrow R/\mathfrak{a}_{mt} \rightarrow 0 \xRightarrow{5.14} \text{vol}(-K_X - tv) \geq (-K_X)^n - t^n \text{vol}(v),$$

$$\beta_X(v) \geq 0 \xrightarrow{5.17, (5.16)} (-K_X)^n \leq \left(1 + \frac{1}{n}\right)^n A_X(v)^n \text{vol}(v).$$

Taking the infimum of the right side over $v \in \text{Val}_{x,X}$ gives (6.5) by Definition 6.1. $\square$

Corollary 6.6 (Global and index bounds).

$$\widehat{\text{vol}}(x, X) \leq n^n \xrightarrow{6.2, (6.5)} (-K_X)^n \leq (n+1)^n, \quad x \in \mathbb{A}^n/G \xrightarrow{(6.5)} (-K_X)^n \leq \frac{(n+1)^n}{|G|},$$

with equality in the first exactly for $X \cong \mathbb{P}^n$.

Proof. [24], [15, §5]. $\square$

Example 6.7 (An unstable weighted plane). For $X = \mathbb{P}(1, 1, 2)$ with singular point $0 \in \mathbb{A}^2/\mu_2$,

$$(-K_X)^2 = \frac{(1+1+2)^2}{1 \cdot 1 \cdot 2} = 8 \xrightarrow{6.6} 8 \leq \frac{3^2}{2} = \frac{9}{2},$$

a contradiction, so X is K -unstable.

Definition 6.8 (The K -moduli functor). Fix n and $V > 0$. The K -moduli functor sends T to the groupoid of flat $\mathbb{Q}$ -Gorenstein families $\mathcal{X} \rightarrow T$ satisfying Kollár's condition of Definition 9.11, namely that every reflexive power $\omega_{\mathcal{X}/T}^{[m]}$ commutes with arbitrary base change, whose geometric fibres are K -semistable $\mathbb{Q}$ -Fano varieties of dimension n and volume V [15, §6], [21, §6].

Theorem 6.9 (The four structural inputs). For $\mathbb{Q}$ -Fano varieties of fixed n and V ,

K -semistable locus bounded $\xrightarrow{[13]}$ one uniform anticanonical embedding, $\text{openness} \xleftrightarrow{5.19}$ finite type,
 K -semistable reduction, unique K -polystable limits $\xrightarrow{2.10}$ properness, $\text{Aut reductive} \xleftrightarrow{5.9}$ good moduli space.

Proof. [13], [14], [7], [8], [15, §6]. $\square$

Theorem 6.10 (K -moduli). There is a proper Artin stack of finite type parameterizing K -semistable $\mathbb{Q}$ -Fano varieties of dimension n and volume V , and it admits a projective good moduli space in the sense of Definition 2.1,

$$\mathcal{M}_{n,V}^{K\text{-ss}} \xrightarrow{\pi} M_{n,V}^K \quad \pi(x_1) = \pi(x_2) \xleftrightarrow{2.4} \overline{\{x_1\}} \cap \overline{\{x_2\}} \neq \emptyset,$$

whose closed points are exactly the K -polystable varieties.

Proof. Each structural input of Theorem 6.9 discharges exactly one hypothesis of the existence criterion of §2.

boundedness, openness $\xrightarrow{6.9}$ $\mathcal{M}_{n,V}^{K\text{-ss}}$ of finite type, K -semistable reduction $\xrightarrow{2.9}$ Θ -reductive, S -complete,

$\text{Aut reductive} \xrightarrow{2.10, 2.5} \pi$ a proper good moduli space with K -polystable closed points

[7], [8], [15, §6]. The closed points are the polystable objects by Definition 2.5, which for this stack are the K -polystable varieties. $\square$

6.1. Cubic Surfaces. The smallest case where the whole architecture can be checked against a classical answer is the cubic surface, where the K -moduli space is a geometric invariant theory quotient.

Theorem 6.11 (Stability equals invariant theory for cubic surfaces). A cubic surface is K -semistable, K -polystable or K -stable if and only if its cubic form is respectively GIT semistable, polystable or stable for PGL_4 , so

$$M_{2,3}^K \cong \mathbb{P}(H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(3)))^{\text{ss}} // \text{PGL}_4,$$

and every smooth cubic surface is K -stable.

Proof. A one-parameter subgroup $\lambda : \mathbb{G}_m \rightarrow \mathrm{PGL}_4$ gives a test configuration, and the Paul–Tian comparison is $\mathrm{Fut}(\mathcal{X}_\lambda, \mathcal{L}_\lambda) = c \mu([X], \lambda)$ with $c > 0$.

Hilbert–Mumford $\xrightarrow{5.7}$ K-stability $\implies$ GIT stability, X GIT polystable $\xrightarrow{5.19}$ a K-stable smoothing $\mathcal{X} \rightarrow C$,

$$\text{properness} \xrightarrow{6.10} \mathcal{X}' \rightarrow C \text{ with K-polystable } X'_0, \quad 3 = (-K_{X'_0})^2 \leq \frac{9}{|G|} \xrightarrow{6.6} 2 \leq |G| \leq 3,$$

A_1, A_2 or $\frac{1}{3}(1, 1)$, the last not smoothable $\xrightarrow{4.14, 4.13} X'_0$ an anticanonically embedded cubic,

X, X'_0 two polystable limits of one punctured family $\xrightarrow{2.4} X \cong X'_0$.

Numerical stability supplied the explicit quotient, the volume bound of Theorem 6.4 limited the admissible singularities, and uniqueness of polystable limits identified the two compactifications [15, §7]. $\square$

Part 4. Birational Geometry¹⁰

7. THE SURFACE MMP

If K_X is not nef, some curve has negative degree against it, and contracting such curves is the whole program. On a surface every birational contraction is the blow-down of Theorem 4.4, so the surface case is the higher-dimensional program with the hard step removed. Throughout (X, Δ) is a projective $\mathbb{Q}$ -factorial klt surface pair, klt in the sense of Definition 8.4 below, and we write

$F := \overline{\text{NE}}(X) \cap D^\perp$ for the face supported by a nef divisor D , cont_R for the contraction of a ray R , $\text{Exc}(f)$ for the exceptional locus of a birational morphism f , $\rho(X)$ the Picard number of §3.

Definition 7.1 (Extremal ray). A one-dimensional face $R \subseteq \overline{\text{NE}}(X)$ is an *extremal ray*, so

$$z_1, z_2 \in \overline{\text{NE}}(X), \quad z_1 + z_2 \in R \implies z_1, z_2 \in R,$$

and a *contraction* is a projective surjection $f: X \rightarrow Y$ with $f_*\mathcal{O}_X = \mathcal{O}_Y$, whose contracted curves span a face [22, §1].

Theorem 7.2 (Cone theorem). *There are countably many rational curves $C_j \subseteq X$ with*

$$(7.3) \quad \overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{(K_X + \Delta) \geq 0} + \sum_j \mathbb{R}_{\geq 0}[C_j], \quad 0 < -(K_X + \Delta) \cdot C_j \leq 4,$$

and for every ample A only finitely many rays lie in $\{(K_X + \Delta + A) < 0\}$, so the rays accumulate only along $(K_X + \Delta)^\perp$; see Figure 9.

Proof. [22, §1], [22, §3]. □

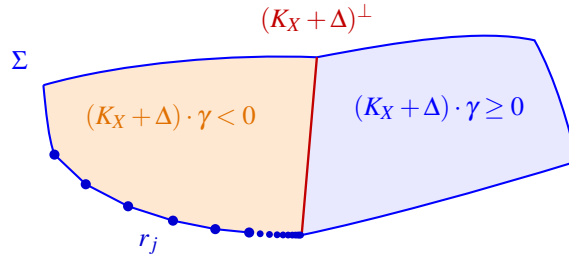


FIGURE 9. For an ample class A , the normalized slice $\Sigma = \overline{\text{NE}}(X) \cap \{\gamma : A \cdot \gamma = 1\}$ turns the negative extremal rays into locally discrete boundary vertices, which may accumulate only at $\Sigma \cap (K_X + \Delta)^\perp$.

Theorem 7.4 (Contraction theorem). *Every $(K_X + \Delta)$ -negative extremal ray R admits a contraction $\text{cont}_R: X \rightarrow Y$ with*

$$[C] \in R \iff \text{cont}_R(C) = \text{pt},$$

and for a nef supporting divisor D of R some multiple of D is base-point free with $D = \text{cont}_R^* A$ for A ample on Y .

Proof. Bend and break produces the rational curves, and the base-point-free theorem converts a supporting divisor into the morphism.

bend and break $\xrightarrow{[22, \S 1]}$ rational curves of bounded anticanonical degree $\xrightarrow{(7.3)}$ the rays are locally discrete,

$$D \text{ rational nef with } \overline{\text{NE}}(X) \cap D^\perp = R \xrightarrow{\text{base-point-free theorem}} |mD| \text{ defines } \text{cont}_R.$$

¹⁰Based on lectures by Kristin DeVleming [3].

The morphism contracts exactly the curves on which D has degree zero, which by construction are the classes in R . $\square$

Theorem 7.5 (Surface MMP). *Repeated contraction of $(K_X + \Delta)$ -negative extremal rays terminates, every step being birational with divisorial exceptional locus, and the output is a minimal model or a Mori fibre space,*

$$\begin{array}{c} X = X_0 \longrightarrow X_1 \longrightarrow \cdots \longrightarrow X_r \\ \swarrow \quad \searrow \\ K_{X_r} + \Delta_r \text{ nef} \quad X_r \rightarrow B, \dim B < 2 \end{array}$$

If every intermediate surface is smooth, each birational step contracts a (-1) -curve.

Proof. On a surface no exceptional locus has codimension two, so every step drops the Picard number and the process is finite.

$$\text{codim Exc}(\text{cont}_R) \geq 2 \xrightarrow{\dim X = 2} \text{Exc}(\text{cont}_R) = \emptyset, \quad \text{each birational step} \xrightarrow{(4.2)} \rho \text{ drops by one,}$$

$$\text{no negative ray remains} \xleftrightarrow{(7.3)} K_{X_r} + \Delta_r \text{ nef}, \quad \dim B < 2 \xrightarrow{7.4} -(K + \Delta) \text{ relatively ample.}$$

The run stops after at most $\rho(X)$ birational steps, and the two displayed terminal cases are exhaustive. $\square$

Theorem 7.6 (Abundance for surfaces). *If $K_X + \Delta$ is nef, then it is semiample, so for sufficiently divisible m it defines the Iitaka morphism*

$$X \xrightarrow{\phi_{|m(K_X + \Delta)|}} Z$$

Proof. [22, §3]. $\square$

7.1. Kodaira Dimension. The Iitaka morphism of Theorem 7.6 has a point, a curve, or a surface as target, and that trichotomy is the Enriques–Kodaira classification.

Definition 7.7 (Plurigeners and Kodaira dimension). For a smooth projective surface the *irregularity*, *geometric genus*, *plurigeners* and *Kodaira dimension* are

$$q(X) := h^1(X, \mathcal{O}_X), \quad p_g(X) := h^0(X, K_X), \quad P_m(X) := h^0(X, mK_X), \quad \kappa(X) \in \{-\infty, 0, 1, 2\},$$

the last being the growth rate of P_m . All four are birational invariants, whereas K_X^2 and $\rho(X)$ change under blow-up by (4.2) [19, §V.6].

Theorem 7.8 (Enriques–Kodaira). *Let X be a smooth minimal projective surface. The Kodaira dimension determines the surface and the endpoint of the program,*

$$\kappa(X) = -\infty \xleftrightarrow{7.5} X \text{ rational or ruled}, \quad \kappa(X) = 0 \xleftrightarrow{7.6} mK_X \sim 0 \text{ for some } m,$$

$$\kappa(X) = 1 \xleftrightarrow{7.6} X \text{ properly elliptic}, \quad \kappa(X) = 2 \xleftrightarrow{??} K_X \text{ big and nef},$$

the $\kappa = 0$ surfaces being K3, abelian, Enriques or bielliptic. For $\kappa(X) \geq 0$ the minimal model is unique, while a ruled surface has infinitely many minimal models.

Proof. [19, §V.6], [22, §1]. $\square$

Proposition 7.9 (Exceptional configurations of the canonical map). *Let X be smooth minimal of general type and $C \subseteq X$ integral with $K_X \cdot C = 0$. Then C is a smooth rational curve with $C^2 = -2$, and each connected configuration of such curves is negative definite with an ADE dual graph.*

Proof. Hodge index forces negativity against the big nef class K_X , and adjunction then leaves a single numerical possibility.

$$K_X \text{ nef and big, } K_X \cdot C = 0 \xrightarrow{3.6} C^2 < 0, \quad 2p_a(C) - 2 = C^2 \xleftrightarrow{3.4} C^2 = -2, \quad p_a(C) = 0,$$

the configuration is K_X -orthogonal $\xrightarrow{3.6}$ negative definite $\xrightarrow{[22, \S 4]}$ ADE dual graph.

The classification of negative-definite graphs of (-2) -classes is exactly the ADE list, which completes the claim. $\square$

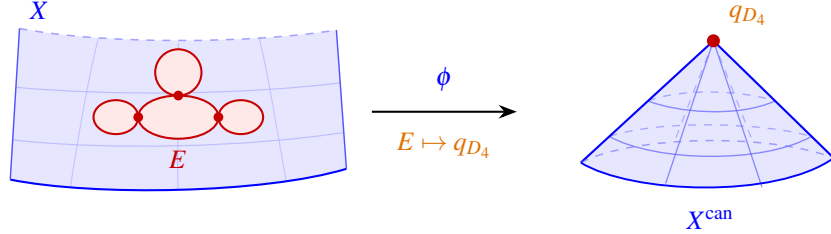


FIGURE 10. The canonical morphism contracts the D_4 configuration $E = E_0 \cup E_1 \cup E_2 \cup E_3$ of (-2) -curves on X to the Du Val point $q_{D_4} \in X^{\text{can}}$.

8. SINGULARITIES OF PAIRS AND FLIPS

The program creates singularities, so its natural objects are pairs, and the admissible singularities are cut out by one numerical invariant. Beyond surfaces the exceptional locus of a birational contraction can have codimension at least two, and repairing that is the flip. Throughout, $\Delta_Y := f_*^{-1}\Delta$ is the strict transform of the boundary, $a(E, X, \Delta)$ is the discrepancy of (8.3), and $\frac{1}{r}(a_1, \dots, a_n)$ denotes the quotient of $\mathbb{A}^n$ by μ_r acting with the listed weights.

Definition 8.1 (Pair and log resolution). A *pair* (X, Δ) is a normal variety X with an effective $\mathbb{Q}$ -divisor Δ such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. A *log resolution* is a projective birational $f: Y \rightarrow X$ with Y smooth and $\Delta_Y \cup \text{Exc}(f)$ a simple normal crossings divisor [22, §2].

Definition 8.2 (Discrepancy and log discrepancy). For a log resolution of (X, Δ) , the *discrepancies* $a(E, X, \Delta)$ are defined by

$$(8.3) \quad K_Y + \Delta_Y = f^*(K_X + \Delta) + \sum_{E \subseteq \text{Exc}(f)} a(E, X, \Delta) E,$$

and the *log discrepancy* and *total discrepancy* are

$$A_{X, \Delta}(E) := 1 + a(E, X, \Delta), \quad \text{discrep}(X, \Delta) := \inf_E a(E, X, \Delta),$$

the infimum over all prime divisors E over X , independent of the resolution because two resolutions are dominated by a third [22, §2]. Retaining Δ_Y on the left of (8.3) is what keeps the boundary visible in the coefficients.

Definition 8.4 (Classes of singularities). The pair (X, Δ) is *terminal*, *canonical*, *klt*, respectively *log canonical*, according as

$$a(E, X, \Delta) > 0, \quad a(E, X, \Delta) \geq 0 \quad (\text{exceptional } E), \quad A_{X, \Delta}(E) > 0, \quad A_{X, \Delta}(E) \geq 0 \quad (\text{all } E \text{ over } X)$$

[22, §2].

Example 8.5 (Blowing up a smooth point). For X smooth of dimension n , $\Delta = \sum d_i D_i$ with $m_i = \text{mult}_p D_i$, and $f = \pi$ as in Construction 4.1,

$$K_Y + \Delta_Y = f^*(K_X + \Delta) + \left(n - 1 - \sum_i d_i m_i\right) E \xrightarrow{8.2} A_{X,\Delta}(E) = n - \sum_i d_i m_i,$$

so $\Delta = 0$ gives $a(E, X, 0) = n - 1 > 0$ and smooth varieties are terminal.

Example 8.6 (Du Val singularities). For the minimal resolution $f: Y \rightarrow X$ of a Du Val surface singularity, as produced by Theorem ??,

$$K_Y = f^* K_X \xrightarrow{8.4} X \text{ canonical but not terminal,}$$

and in dimension two terminal singularities are smooth points [22, §4].

Definition 8.7 (Log canonical threshold). For a log canonical (X, Δ) and an effective $\mathbb{Q}$ -Cartier D , the *log canonical threshold* is

$$\text{lct}(X, \Delta; D) := \sup \{t \in \mathbb{Q}_{\geq 0} \mid (X, \Delta + tD) \text{ log canonical}\}, \quad \text{lct}(X, D) := \text{lct}(X, 0; D)$$

[21, §8], [22, §2].

Proposition 8.8 (Computation on a resolution). *For a log resolution of $(X, \Delta + D)$,*

$$\text{lct}(X, \Delta; D) = \min_E \frac{A_{X,\Delta}(E)}{\text{ord}_E(f^* D)},$$

the minimum over the finitely many $E \subseteq Y$ with $\text{ord}_E(f^ D) > 0$.*

Proof. On one fixed resolution every discrepancy is an affine function of the coefficient t , so the threshold is computed by finitely many linear inequalities.

$$K_Y + \Delta_Y = f^*(K_X + \Delta + tD) + \sum_E (a(E, X, \Delta) - t \text{ord}_E(f^* D)) E \xleftrightarrow{(8.3)} \text{coefficients affine in } t,$$

$$(X, \Delta + tD) \text{ log canonical} \xleftrightarrow{8.4} A_{X,\Delta}(E) \geq t \text{ord}_E(f^* D) \quad (\text{all } E).$$

The largest admissible t is the displayed minimum, by Definition 8.7. $\square$

Example 8.9 (Plane curves). For $X = \mathbb{A}^2$ and D with an ordinary point of multiplicity m , Example 8.5 gives $A_{X,0}(E) = 2$ and $\text{ord}_E(\pi^* D) = m$, so

$$\text{lct}(\mathbb{A}^2, D) \leq \frac{2}{m}, \quad \text{lct}(\mathbb{A}^2, (xy)) = 1, \quad \text{lct}(\mathbb{A}^2, (x^2 - y^3)) = \frac{5}{6}, \quad \text{lct}(\mathbb{A}^2, (x^a + y^b)) = \frac{1}{a} + \frac{1}{b}.$$

8.1. Beyond Surfaces. On a $\mathbb{Q}$ -factorial surface a birational exceptional locus is a divisor, so Theorem 7.5 needs no flips. In dimension three both the singularities and the small contractions are unavoidable.

Example 8.10 (A divisorial contraction with fractional discrepancy). Let $X = \mathbb{P}_{\mathbb{P}^2}(\mathcal{O} \oplus \mathcal{O}(2))$, let $D \cong \mathbb{P}^2$ be the section with $N_{D/X} \cong \mathcal{O}_{\mathbb{P}^2}(-2)$, and let $L \subseteq D$ be a line.

$$D \cdot L = -2, \quad (K_X + D)|_D = K_D \xrightarrow{3.4} K_X \cdot L = K_D \cdot L - D \cdot L = -3 + 2 = -1,$$

so $\mathbb{R}_{\geq 0}[L]$ is K_X -negative and its contraction f sends D to a point. Writing $K_X = f^* K_Y + aD$,

$$-1 = a \cdot (-2) \xrightarrow{a = \frac{1}{2} \notin \mathbb{Z}} Y \text{ is singular,}$$

the discrepancy $1/2$ matching a terminal quotient singularity of type $\frac{1}{2}(1, 1, 1)$; see Figure 11.

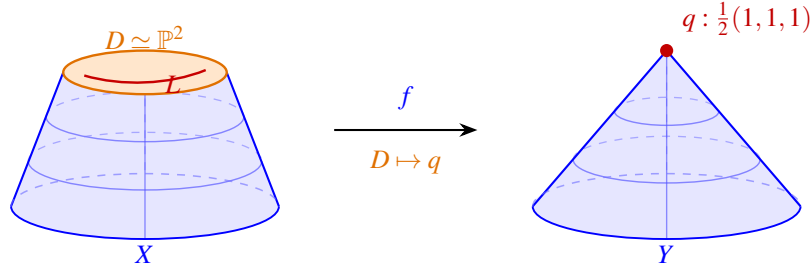


FIGURE 11. Locally, the contraction collapses the exceptional section $D \simeq \mathbb{P}^2$, with $N_{D/X} \simeq \mathcal{O}_{\mathbb{P}^2}(-2)$, to the vertex q of the quotient cone of type $\frac{1}{2}(1, 1, 1)$.

Definition 8.11 (Small contraction, flop and flip). A birational contraction $f: X \rightarrow Y$ is *small* when $\text{codim}_X \text{Exc}(f) \geq 2$. If $K_X \cdot C = 0$ for every contracted curve, a replacement of $\text{Exc}(f)$ by another small resolution is a *flop*; if $K_X \cdot C < 0$, the replacement is a *flip*,

$$\begin{array}{ccc}
 X & \xrightarrow{\text{flip}} & X^+ \\
 \searrow f & & \swarrow f^+ \\
 & Y &
 \end{array}
 \quad - (K_X + \Delta) \text{ } f\text{-ample}, \quad K_{X^+} + \Delta^+ \text{ } f^+\text{-ample}.$$

Example 8.12 (The Atiyah flop). Let Y be the threefold ordinary double point, the affine cone over $\mathbb{P}^1 \times \mathbb{P}^1$. Blowing up the vertex gives exceptional divisor $E \cong \mathbb{P}^1 \times \mathbb{P}^1$, and contracting either ruling gives a small resolution,

$$\text{Exc}(f) \cong \mathbb{P}^1, \quad K_X \cdot \text{Exc}(f) = 0 \xrightarrow{8.11} X \dashrightarrow X^+ \text{ is a flop},$$

as in Figure 12.

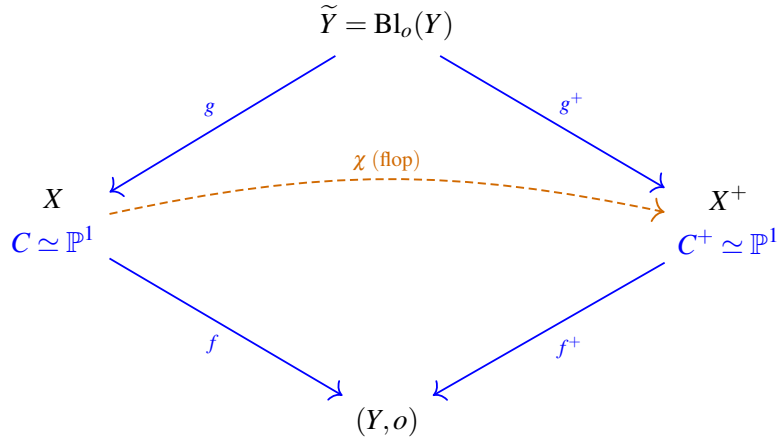


FIGURE 12. The blow-up of the cone vertex has exceptional divisor $E \simeq \mathbb{P}^1 \times \mathbb{P}^1$; contracting its two rulings gives the small resolutions X and X^+ related by the Atiyah flop.

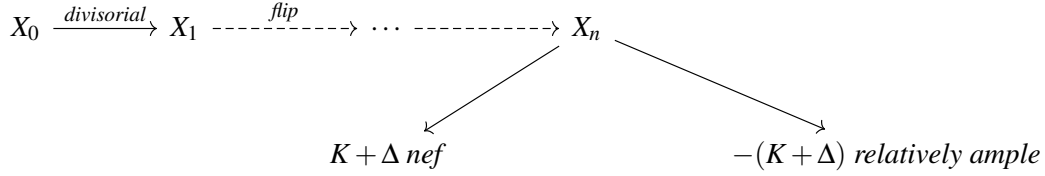
Theorem 8.13 (Existence and uniqueness of flips). *Flips of klt pairs exist, and existence forces uniqueness,*

$$(8.14) \quad X^+ = \operatorname{Proj}_Y \bigoplus_{m \geq 0} f_* \mathcal{O}_X(\lfloor m(K_X + \Delta) \rfloor),$$

the relative log canonical ring being finitely generated.

Proof. [22, §6], [21, §1]. □

Theorem 8.15 (Minimal models and finite generation). *For a projective klt pair (X, Δ) with $K_X + \Delta$ big, a log minimal model exists, the log canonical ring is finitely generated, and the program alternates divisorial contractions with flips before reaching one of two endpoints,*



Proof. [21, §1], [22, §3]. □

Definition 8.16 (Log canonical model). When $K_X + \Delta$ is big, the *log canonical model* of (X, Δ) is

$$X^{\text{can}} := \operatorname{Proj} \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(\lfloor m(K_X + \Delta) \rfloor)),$$

finitely generated by Theorem 8.15, unique in the birational class, and carrying an ample $K + \Delta$. This is the model by which §9 defines stability [21, §1], [12, §1].

Part 5. KSBA Moduli¹¹

9. STABLE PAIRS

A limit of normal varieties need not be normal, so the class of objects in §7 is too small to be proper. This section enlarges it in exactly one direction, allowing double loci in codimension one and nothing worse, and then shows that the enlarged objects are still rigid enough to have finite automorphism groups. Throughout (X, Δ) is projective with $\Delta = \sum a_i D_i$ effective and $K_X + \Delta$ $\mathbb{Q}$ -Cartier, and we write

$v: \bar{X} \rightarrow X$ for the normalization, $D \subseteq X$, $\bar{D} \subseteq \bar{X}$ for the conductor divisors, $\bar{D}^v$ the normalization of $\bar{D}$, $\tau: \bar{D}^v \rightarrow \bar{D}^v$ the gluing involution, $\mathcal{F}^{[m]} := (\mathcal{F}^{\otimes m})^{\vee\vee}$ the reflexive powers.

Definition 9.1 (Demi-normal scheme). A reduced scheme X is *demi-normal* when it satisfies Serre's condition S_2 and every codimension-one point is either regular or an ordinary double point [21, §5]. Condition S_2 is the algebraic Hartogs principle, so for $j: U \hookrightarrow X$ with $\text{codim}(X \setminus U) \geq 2$ and $\mathcal{F}$ rank-one reflexive,

$$\mathcal{F} \cong j_*(\mathcal{F}|_U).$$

Construction 9.2 (Conductor and gluing involution). For X demi-normal the *conductor ideal* $\mathcal{H}om_{\mathcal{O}_X}(v_*\mathcal{O}_{\bar{X}}, \mathcal{O}_X)$ cuts out reduced divisors $D \subseteq X$ and $\bar{D} \subseteq \bar{X}$, the map $\bar{D} \rightarrow D$ is generically two-to-one, and the two branches are exchanged by an involution τ with $\tau^2 = \text{id}$,

$$\bar{D}^v \xrightarrow[l \circ \tau]{l} \bar{X} \xrightarrow{v} X = \bar{X}/R(\tau)$$

the quotient existing precisely when the relation generated by the two maps is finite [21, §5]. Figure 13 shows the local picture along D .

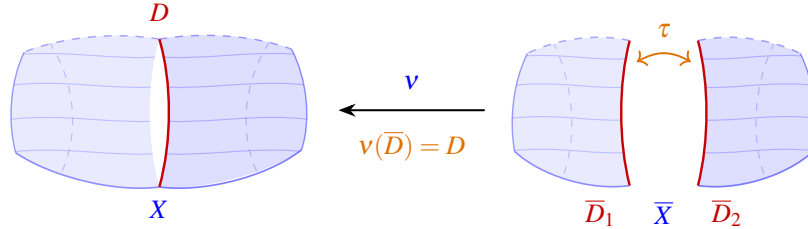


FIGURE 13. Normalization separates the two branches of the double locus, while the gluing data survives as the involution τ exchanging the two components of the normalized conductor $\bar{D}$.

Definition 9.3 (Semi-log canonical pair). Let X be demi-normal, let Δ contain no component of D , and let $\bar{\Delta} := v_*^{-1}\Delta$. Then

$$(9.4) \quad v^*(K_X + \Delta) = K_{\bar{X}} + \bar{D} + \bar{\Delta},$$

and (X, Δ) is *semi-log canonical* when $(\bar{X}, \bar{D} + \bar{\Delta})$ is log canonical in the sense of Definition 8.4. It is a *stable pair* when in addition $K_X + \Delta$ is ample [21, §5], [12, §1].

Example 9.5 (Curves). For a reduced connected curve,

$$\text{demi-normal} \xleftrightarrow{9.1} \text{nodal}, \quad v^*K_X = K_{\bar{X}} + \bar{D} \xrightarrow{(9.4), 9.3} \text{stable pairs are stable curves},$$

$\bar{D}$ being the two preimages of each node, and adding a boundary recovers the weighted stability of §11 [20], [21, §5].

¹¹Based on lectures by Dori Bejleri [2].

Definition 9.6 (Different). For $(X, S + \Delta)$ with S reduced, no component of S inside $\text{Supp}\Delta$, and $K_X + S + \Delta$ $\mathbb{Q}$ -Cartier, the *different* is the unique effective $\mathbb{Q}$ -divisor $\text{Diff}_{S^\vee}(\Delta)$ on $S^\vee$ with

$$(9.7) \quad (K_X + S + \Delta)|_{S^\vee} = K_{S^\vee} + \text{Diff}_{S^\vee}(\Delta),$$

so (9.7) is Theorem 3.4 with the correction forced by the singularities of S [21, §4].

Theorem 9.8 (Kollár gluing). *Normalization and gluing are inverse equivalences between stable pairs (X, Δ) and triples $(\bar{X}, \bar{D} + \bar{\Delta}, \tau)$ with $(\bar{X}, \bar{D} + \bar{\Delta})$ a log canonical pair whose log canonical class is ample, τ an involution of $\bar{D}^\vee$ exchanging the two generic branches, and*

$$\tau^* \text{Diff}_{\bar{D}^\vee}(\bar{\Delta}) = \text{Diff}_{\bar{D}^\vee}(\bar{\Delta}),$$

the relation of Construction 9.2 being finite.

Proof. The pluricanonical forms on X are exactly the τ -invariant forms on the normalization, so the different is the entire gluing obstruction.

$$\tau \text{ preserves the different} \xrightarrow{(9.7), (9.4)} \omega_X^{[m]}(m\bar{D} + \lfloor m\bar{\Delta} \rfloor)^\tau = v_* \omega_X^{[m]}(\lfloor m\Delta \rfloor),$$

$$R(\tau) \text{ finite} \xrightarrow{[21, \S 9]} X = \bar{X}/R(\tau) \text{ exists} \xleftrightarrow{9.3} (X, \Delta) \text{ stable.}$$

Descent of the invariant algebra reconstructs X from the triple, and normalizing recovers the triple from X . $\square$

Example 9.9 (Two components glued along a curve). For normal surfaces X_1, X_2 containing isomorphic smooth curves $D_i \subseteq X_i$ and an isomorphism $\tau : D_1 \rightarrow D_2$,

$(X_i, D_i + \Delta_i)$ log canonical, $\tau^* \text{Diff}_{D_2}(\Delta_2) = \text{Diff}_{D_1}(\Delta_1)$, $K_{X_i} + D_i + \Delta_i$ ample $\xrightarrow{9.8} X_1 \cup_D X_2$ stable, the surface analogue of gluing the two preimages of a node in Example 9.5; see Figure 14.

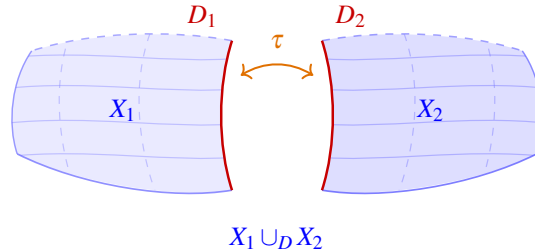


FIGURE 14. Gluing the normal surface pairs (X_1, D_1) and (X_2, D_2) through $\tau : D_1 \xrightarrow{\sim} D_2$, with matching differentials, produces the nonnormal semi-log canonical surface $X_1 \cup_D X_2$.

Theorem 9.10 (Finiteness of automorphisms). *For a stable pair (X, Δ) the group $\text{Aut}(X, \Delta)$ is finite and reduced.*

Proof. A positive-dimensional automorphism group would supply a global vector field preserving the ample log canonical class, and ampleness forbids it.

$$\text{Aut}(X, \Delta) \text{ positive dimensional} \xrightarrow{9.3} \text{a vector field preserves the ample } K_X + \Delta,$$

$$K_X + \Delta \text{ ample} \xrightarrow{[21, \S 5]} H^0(X, T_X(-\log \Delta)) = 0 \xrightarrow{\text{characteristic } 0} \text{Aut}(X, \Delta) \text{ finite and reduced.}$$

The vanishing contradicts the vector field, and reducedness is automatic in characteristic zero. $\square$

9.1. Kollár's Condition. Fibrewise stability is not deformation invariant, because reflexive pluricanonical sheaves need not restrict correctly to fibres. The condition below is the family-level replacement for $\mathbb{Q}$ -Cartierness on a single fibre.

Definition 9.11 (Stable family). A *stable family* is a flat projective $f: (\mathcal{X}, \Delta) \rightarrow T$ with every fibre (X_t, Δ_t) semi-log canonical, $K_{\mathcal{X}/T} + \Delta$ relatively ample, and, for every $m \in \mathbb{Z}$, the reflexive sheaf $\omega_{\mathcal{X}/T}^{[m]}(\lfloor m\Delta \rfloor)$ flat over T and compatible with arbitrary base change,

$$\begin{array}{ccc} \omega_{\mathcal{X}'/T'}^{[m]}(\lfloor m\Delta' \rfloor) & \xrightarrow{\cong} & g^* \omega_{\mathcal{X}/T}^{[m]}(\lfloor m\Delta \rfloor) \\ \downarrow & & \downarrow \\ T' & \xrightarrow{g} & T \end{array}$$

the last requirement being *Kollár's condition* [21, §3], [12, §2].

Definition 9.12 (Canonical covering stack). Let r be the *index* of $K_X + \Delta$ at x , the least $r > 0$ with $r(K_X + \Delta)$ Cartier near x . Locally the *canonical covering stack* is

$$\mathfrak{X} := \left[\operatorname{Spec}_X \bigoplus_{i=0}^{r-1} \mathcal{O}_X(i(K_X + \Delta)) / \mu_r \right] \xrightarrow{1.8} \omega_{\mathfrak{X}}(\Delta) \text{ is a line bundle,}$$

so a deformation is $\mathbb{Q}$ -Gorenstein exactly when $\mathfrak{X}$ deforms [21, §3], [11, §6].

Proposition 9.13 (Deformations of a stable pair). *The $\mathbb{Q}$ -Gorenstein deformation theory of (X, Δ) is controlled by the cotangent complex of $\mathfrak{X}$,*

$$T^1 \cong \operatorname{Ext}^1(L_{\mathfrak{X}}, \mathcal{O}_{\mathfrak{X}}), \quad T^2 \subseteq \operatorname{Ext}^2(L_{\mathfrak{X}}, \mathcal{O}_{\mathfrak{X}}),$$

and locally trivial deformations form a proper subspace of T^1 .

Proof. [21, §3], [11, §6]. □

10. STABLE REPLACEMENT AND PROPERNESS

Properness is the statement that a punctured family of stable pairs has a limit, and separatedness that the limit is unique. Both are proved by replacing the family with the relative log canonical model of Definition 8.16, so the minimal model program of §7 is what makes the moduli space compact. Throughout R is the discrete valuation ring of §2, with fraction field K , residue field κ , and finite extensions $R \subseteq R'$ with fraction fields $K \subseteq K'$.

Construction 10.1 (Stable replacement). Given a stable family over $\operatorname{Spec} K$, pass to a finite extension $R \subseteq R'$, choose a $\mathbb{Q}$ -factorial dlt model $(Y, \Delta_Y + Y_0) \rightarrow \operatorname{Spec} R'$ with reduced central fibre, and set

$$(10.2) \quad (\mathcal{X}, \Delta_{\mathcal{X}} + X_0) := \operatorname{Proj}_{R'} \bigoplus_{m \geq 0} f_* \mathcal{O}_Y(\lfloor m(K_Y + \Delta_Y + Y_0) \rfloor),$$

finitely generated by Theorem 8.13. The finite extension is unavoidable, since monodromy around the puncture must be killed before a semistable model exists [21, §2], [12, §2].

Theorem 10.3 (Stable reduction). *The pair (10.2) is a stable family and its generic fibre is the given one, so the dashed extension exists after finite base change and is unique up to unique isomorphism,*

$$\begin{array}{ccc} \operatorname{Spec} K' & \xrightarrow{\text{punctured family}} & \mathcal{M}^{\text{KSBA}} \\ \downarrow & \nearrow \exists! & \\ \operatorname{Spec} R' & & \end{array}$$

Proof. Adjunction on the central fibre turns log canonicity of the total space into semi-log canonicity of X_0 .

$$(K_{\mathcal{X}} + \Delta_{\mathcal{X}} + X_0)|_{X_0} \xLeftrightarrow{(9.7)} K_{X_0} + \text{Diff}_{X_0}(\Delta_{\mathcal{X}}) \xrightarrow{9.3, [21, \S 4]} (X_0, \Delta_0) \text{ semi-log canonical,}$$

$$K_{\mathcal{X}} + \Delta_{\mathcal{X}} + X_0 \text{ relatively ample} \xrightarrow{(10.2), 9.8} (\mathcal{X}, \Delta_{\mathcal{X}}) \rightarrow \text{Spec } R' \text{ stable,}$$

two fillings share a common resolution $W \xrightarrow{8.16}$ the same relative log canonical algebra.

The relative Proj of that common algebra is the unique filling, which is the dashed arrow. $\square$

Proposition 10.4 (Separatedness). *Stable families $(\mathcal{X}_1, \Delta_1)$ and $(\mathcal{X}_2, \Delta_2)$ over $\text{Spec } R$ that agree over $\text{Spec } K$ are isomorphic,*

$$\begin{array}{ccc} & W & \\ g_1 \swarrow & & \searrow g_2 \\ \mathcal{X}_1 & \xrightarrow{\cong} & \mathcal{X}_2 \\ & \searrow & \swarrow \\ & \text{Spec } R & \end{array}$$

Proof. Both families compute the same ample model of a common resolution, and ample models are unique.

$$\mathcal{X}_i = \text{Proj}_R \bigoplus_{m \geq 0} (g_i)_* \mathcal{O}_W(\lfloor m(K_W + \Theta) \rfloor) \xLeftrightarrow{8.16} \mathcal{X}_i \text{ is the ample model of } W,$$

discrepancies and the generic identification agree $\xrightarrow{(10.2)} \mathcal{X}_1 \cong \mathcal{X}_2$.

The generic isomorphism extends because both sides are Proj of one graded algebra. $\square$

Theorem 10.5 (Boundedness). *Fix n , a volume $v > 0$, and a set $I \subseteq (0, 1] \cap \mathbb{Q}$ satisfying the descending chain condition. There is $m = m(n, v, I) > 0$ such that $m(K_X + \Delta)$ is Cartier and very ample with fixed Hilbert polynomial for every stable pair with those data, so $\lfloor m(K_X + \Delta) \rfloor$ embeds every such pair in one $\mathbb{P}^N$ and*

$$[U/\text{PGL}_{N+1}] \leftarrow U \hookrightarrow \text{Hilb}(\mathbb{P}^N)$$

with U the locally closed locus cut out by the semi-log canonical, coefficient and Kollár conditions.

Proof. [21, §6], [13]. $\square$

Theorem 10.6 (KSBA moduli). *For fixed n , v and I as in Theorem 10.5, the stable pairs of dimension n , volume v and coefficients in I form a proper Deligne–Mumford stack $\mathcal{M}_{n,v,I}^{\text{KSBA}}$ of finite type whose coarse moduli space is projective.*

Proof. Every hypothesis in the definition of a stable pair funds exactly one line below.

$$\text{Aut}(X, \Delta) \text{ finite and reduced} \xrightarrow{9.10} \text{unramified diagonal,} \quad \text{Theorem 10.5} \xrightarrow{1.10} \text{finite type,}$$

$$\text{Theorem 10.3, Proposition 10.4} \xrightarrow{\text{valuative criterion}} \text{proper,} \quad \text{semipositivity of } f_* \omega_{\mathcal{X}/T}^{[m]} \xrightarrow{[21, \S 7]} \text{projective.}$$

Semi-log canonicity supplied the objects, ampleness removed continuous automorphisms, Kollár’s condition made the functor deformation invariant, and the relative log canonical model with its uniqueness supplied properness; varying the coefficients I with the rest fixed is the subject of §11 [12, §1], [16, §1]. $\square$

Part 6. Wall Crossings¹²

11. WEIGHTED STABLE PAIRS

Changing the weights changes two geometric tests: which markings may collide and which components remain visible to the log canonical bundle. Equality in either test is a wall. Weighted curves make both mechanisms explicit.

Fix $g, n \geq 0$, $\mathbf{a} = (a_1, \dots, a_n) \in (0, 1]^n \cap \mathbb{Q}^n$, and write

$$\mathcal{C}_{g,n} := \left\{ \mathbf{a} : 2g - 2 + \sum_i a_i > 0 \right\}, \quad \omega_C(\mathbf{a}) := \omega_C\left(\sum_i a_i p_i\right).$$

Definition 11.1 (Weighted stable curve). A connected nodal pointed curve $(C, p_1, \dots, p_n)$ is $\mathbf{a}$ -stable when the markings lie in C_{sm} and

$$p_{i_1} = \dots = p_{i_s} \implies \sum_{j=1}^s a_{i_j} \leq 1, \quad \omega_C(\mathbf{a}) \text{ ample}$$

[20], [16, §1].

Proposition 11.2 (Componentwise stability). For $E \subseteq C$ irreducible, let n_E be the number of nodal branches on its normalization. One has

$$(11.3) \quad \deg_E \omega_C(\mathbf{a}) = 2g(E) - 2 + n_E + \sum_{p_i \in E} a_i.$$

Consequently,

$$\omega_C(\mathbf{a}) \text{ ample} \xLeftrightarrow{11.3} 2g(E) - 2 + n_E + \sum_{p_i \in E} a_i > 0 \quad (E \subseteq C).$$

In particular,

$$E \text{ a rational tail} \xLeftrightarrow{11.3} \sum_{p_i \in E} a_i > 1, \quad E \text{ a rational bridge} \xLeftrightarrow{11.3} \sum_{p_i \in E} a_i > 0.$$

Proof. The normalization of E separates its nodal branches.

$$v^* \omega_C(\mathbf{a})|_E = \omega_{\tilde{E}} \left(\sum_{q \in v^{-1}(C_{\text{sing}})} q + \sum_{p_i \in E} a_i \tilde{p}_i \right) \xRightarrow{3.4} (11.3).$$

□

Theorem 11.4 (Hassett). Every $\mathbf{a} \in \mathcal{C}_{g,n}$ determines a smooth proper Deligne–Mumford stack $\overline{\mathcal{M}}_{g,\mathbf{a}}$ with projective coarse space. The walls are contained in

$$\mathcal{W}_I := \left\{ \mathbf{a} : \sum_{i \in I} a_i = 1 \right\}, \quad I \subseteq \{1, \dots, n\}, \quad |I| \geq 2.$$

Weights in one chamber give the same underlying marked universal curve, while $\mathbf{b} \leq \mathbf{a}$ in $\mathcal{C}_{g,n}$ induces

$$\overline{\mathcal{M}}_{g,\mathbf{a}} \xrightarrow{\rho_{\mathbf{b},\mathbf{a}}} \overline{\mathcal{M}}_{g,\mathbf{b}}.$$

Proof. This is Hassett’s weighted stabilization.

$$\text{component inequalities} \xRightarrow{11.2} \text{finite chambers} \xRightarrow{[20], [16, \S 1]} \overline{\mathcal{M}}_{g,\mathbf{a}}, \rho_{\mathbf{b},\mathbf{a}}.$$

□

¹²Based on lectures by Kenneth Ascher [1].

Construction 11.5 (Reduction morphism). For an $\mathbf{a}$ -stable family $\pi: (\mathcal{C}, \sigma_1, \dots, \sigma_n) \rightarrow T$ and $\mathbf{b} \leq \mathbf{a}$, write $L_{\mathbf{b}} := K_{\mathcal{C}/T} + \sum_i b_i \sigma_i$ and

$$(11.6) \quad \mathcal{C}_{\mathbf{b}} := \text{Proj}_T \bigoplus_{m \geq 0} \pi_* \mathcal{O}_{\mathcal{C}}(\lfloor mL_{\mathbf{b}} \rfloor).$$

For a rational tail E ,

$$\sum_{p_i \in E} b_i \leq 1 \xLeftrightarrow{11.3} \deg_E L_{\mathbf{b}} \leq 0 \xRightarrow{8.16} E \text{ is contracted.}$$

Iterating contracts the maximal $\mathbf{b}$ -unstable tails; their markings become the collisions allowed by Definition 11.1 [20].

Example 11.7 (A four-point wall). For $g = 0$ and $\mathbf{a} = (1, 1, \varepsilon, \varepsilon)$,

$$2\varepsilon > 0 \xLeftrightarrow{11.1} \mathbf{a} \in \mathcal{C}_{0,4}, \quad 2\varepsilon \leq 1 \xLeftrightarrow{11.2} \deg_E \omega_{\mathcal{C}}(\mathbf{a}) \leq 0$$

for the rational tail E carrying p_3, p_4 . At $2\varepsilon = 1$, the tail disappears and its two markings become coincident.

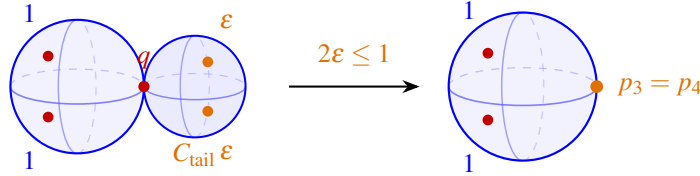


FIGURE 15. Crossing the Hassett wall $2\varepsilon = 1$ contracts the rational tail carrying the two light markings, which then coincide at its former attachment point on the surviving component.

Example 11.8 (Losev–Manin space). For $0 < \varepsilon \leq 1/n$, the weights $(1, 1, \varepsilon, \dots, \varepsilon)$ force chains of rational curves with the two heavy markings at opposite ends and a light marking on every component:

$$\text{rational bridge } E \xLeftrightarrow{11.2} \sum_{p_i \in E} \varepsilon > 0.$$

The resulting Hassett space is the toric permutohedral Losev–Manin space [20].

11.1. Chamber Decompositions. For higher-dimensional pairs the same two tests move linearly in the coefficients. On a fixed log resolution every log discrepancy is an affine function of $\mathbf{a}$, and against a fixed curve class the degree of $K_X + \Delta(\mathbf{a})$ is affine as well, so each potential wall is a rational hyperplane. What is not formal is finiteness, and that is exactly what finite generation of the adjoint ring buys. Throughout, X is a normal projective $\mathbb{Q}$ -Gorenstein variety, $D_1, \dots, D_r$ are prime $\mathbb{Q}$ -Cartier divisors, $P \subseteq [0, 1]^r$ is a rational polytope, and $\Delta(\mathbf{a}) := \sum_i a_i D_i$.

Definition 11.9 (Coefficient polytope and adjoint ring). The *log canonical region* and the *pseudo-effective region* are

$$\mathcal{L} := \{\mathbf{a} \in P : (X, \Delta(\mathbf{a})) \text{ log canonical}\}, \quad \mathcal{E} := \{\mathbf{a} \in \mathcal{L} : K_X + \Delta(\mathbf{a}) \text{ pseudo-effective}\},$$

and for rational boundaries $\Delta_1, \dots, \Delta_s$ spanning the region the *adjoint multigraded ring* is

$$(11.10) \quad R(X; K_X + \Delta_1, \dots, K_X + \Delta_s) := \bigoplus_{\mathbf{m} \in \mathbb{N}^s} H^0\left(X, \mathcal{O}_X\left(\left\lfloor \sum_j m_j (K_X + \Delta_j) \right\rfloor\right)\right).$$

Proposition 11.11 (Linearity of the walls). *Let $f: Y \rightarrow X$ be a log resolution of $(X, \sum_i D_i)$, write $K_Y + B_Y(\mathbf{a}) = f^*(K_X + \Delta(\mathbf{a}))$, and fix a curve class R . Then*

$$\text{coeff}_F B_Y(\mathbf{a}) = 1 - A_{X,0}(F) + \sum_i a_i \text{ord}_F(f^* D_i), \quad (K_X + \Delta(\mathbf{a})) \cdot R = K_X \cdot R + \sum_i a_i (D_i \cdot R),$$

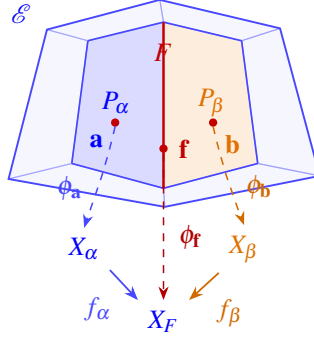
both affine linear in $\mathbf{a}$; in particular $\mathcal{L}$ is a rational polytope.

Proof. One resolution computes every discrepancy at once, and intersection numbers are linear in the divisor.

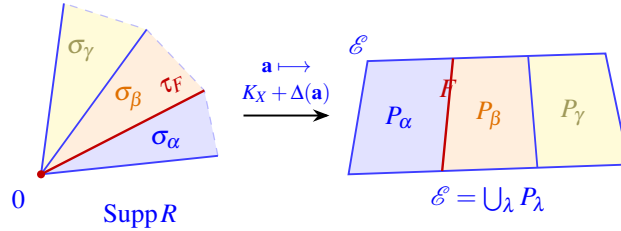
$$K_Y + B_Y(\mathbf{a}) = f^*(K_X + \Delta(\mathbf{a})) \xLeftrightarrow{(8.3)} A_{X,\Delta(\mathbf{a})}(F) = 1 - \text{coeff}_F B_Y(\mathbf{a}) \xRightarrow{8.8} \mathcal{L} = P \cap \bigcap_F \{ \text{coeff}_F B_Y(\mathbf{a}) \leq 1 \},$$

with F running over the finitely many divisors on Y , so both wall families are finitely many rational hyperplanes by the logic above. $\square$

Theorem 11.12 (Geography of log canonical models). *Under finite generation, $\mathcal{E}$ decomposes into rational ample-model chambers separated by contraction walls:*



Proof. Finitely many generator degrees of (11.10) cut a finite fan in degree space, with one Veronese subring and hence one Proj per cone by Definition 8.16.



The slice carries cones to chambers and faces to walls, so the smaller Veronese subring of a face gives the contraction $f_\alpha: X_\alpha \rightarrow X_F$, as in Figure 16, by the logic above [5, Corollary 1.1.5], [12, §2]. $\square$

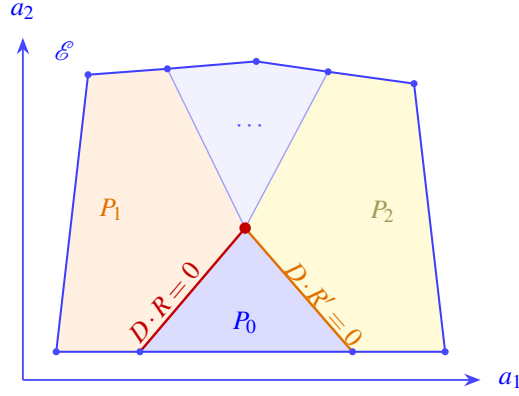


FIGURE 16. The adjoint multigraded ring cuts the pseudo-effective slice into finitely many rational chambers separated by numerical walls.

Definition 11.13 (The three wall types). Let R be the class of the curves contracted at the wall. The wall is *divisorial*, *flipping*, respectively a *support wall*, according as

$$R \text{ sweeps a divisor} \xrightarrow{7.4} \text{codimExc}(R) \geq 2 \xrightarrow{8.11} a_i = 0 \text{ for some } i.$$

A boundary face of $\mathcal{E}$ may instead produce a fibre-type model rather than an internal birational wall.

12. WALL CROSSING MORPHISMS

Remark 12.1 (The Hassett–Keel program). For the boundary $\delta \subseteq \overline{\mathcal{M}}_g$, define

$$\overline{\mathcal{M}}_g(\alpha) := \text{Proj} \bigoplus_{m \geq 0} H^0 \left(\overline{\mathcal{M}}_g, \mathcal{O}_{\overline{\mathcal{M}}_g} (\lfloor m(K_{\overline{\mathcal{M}}_g} + \alpha\delta) \rfloor) \right).$$

On every finitely generated big interval,

$$\alpha \text{ decreases} \xrightarrow{11.12, 8.16} \text{a finite chain of birational models } \overline{\mathcal{M}}_g(\alpha).$$

The Hassett–Keel program asks for the modular meaning of these models and their wall contractions; it does not assert that every genus has already been completely classified [12, §4], [16, §4].

REFERENCES

- [1] K. Ascher, *Algebraic Surfaces and Wall Crossings*. Lecture notes, Moduli of Varieties.
- [2] D. Bejleri, *Stacks and Moduli and KSBA Moduli*. Lecture notes, Moduli of Varieties.
- [3] K. DeVleming, *K-Stability and Birational Geometry*. Lecture notes, Moduli of Varieties.
- [4] K. Ascher, D. Bejleri, G. Inchiostro, and Z. Patakfalvi, *Wall crossing for moduli of stable log pairs*, Ann. of Math. (2) 198 (2023), 825–866.
- [5] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. 23 (2010), 405–468.
- [6] F. Meng and Z. Zhuang, *MMP for locally stable families and wall crossing for moduli of stable pairs*, arXiv:2311.01319.
- [7] J. Alper, H. Blum, D. Halpern-Leistner, and C. Xu, *Reductivity of the automorphism group of K-polystable Fano varieties*, Invent. Math. 222 (2020), 995–1032.
- [8] J. Alper, D. Halpern-Leistner, and J. Heinloth, *Existence of moduli spaces for algebraic stacks*, Invent. Math. 234 (2023), 949–1038.
- [9] J. Alper, J. Hall, and D. Rydh, *A Luna étale slice theorem for algebraic stacks*, Ann. of Math. 191 (2020), 675–738.
- [10] V. Alexeev, *Moduli spaces $M_{g,n}(W)$ for surfaces*, in Higher-dimensional complex varieties, de Gruyter, Berlin, 1996, 1–22.
- [11] J. Alper, *Stacks and moduli*, Graduate Texts in Mathematics, Springer, Cham, Switzerland, 2027, in press.
- [12] D. Bejleri, *Moduli of varieties of general type and wall-crossing*, preprint, University of Maryland–College Park, 2025.
- [13] C. Birkar, *Anti-pluricanonical systems on Fano varieties*, Ann. of Math. 190 (2019), 345–463.
- [14] H. Blum and M. Jonsson, *Thresholds, valuations, and K-stability*, Adv. Math. 365 (2020), 107062.

- [15] K. DeVleming, *On moduli of Fano varieties: an introduction to K -stability and K -moduli*, preprint, University of California–San Diego, 2026.
- [16] K. DeVleming, *A guide to wall crossing for moduli of varieties*, preprint, University of California–San Diego, 2026.
- [17] K. Fujita, *A valuative criterion for uniform K -stability of $\mathbb{Q}$ -Fano varieties*, J. Reine Angew. Math. 751 (2019), 309–338.
- [18] P. Hacking, *Compact moduli of plane curves*, Duke Math. J. 124 (2004), 213–257.
- [19] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics 52, Springer, New York, NY, 1977.
- [20] B. Hassett, *Moduli spaces of weighted pointed stable curves*, Adv. Math. 173 (2003), 316–352.
- [21] J. Kollár, *Families of varieties of general type*, Cambridge Tracts in Mathematics 231, Cambridge University Press, Cambridge, 2023.
- [22] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Mathematics 134, Cambridge University Press, Cambridge, 1998.
- [23] J. Kollár and N. I. Shepherd-Barron, *Threefolds and deformations of surface singularities*, Invent. Math. 91 (1988), 299–338.
- [24] C. Li, *Minimizing normalized volumes of valuations*, Math. Z. 289 (2018), 491–513.
- [25] C. Li and C. Xu, *Special test configuration and K -stability of Fano varieties*, Ann. of Math. 180 (2014), 197–232.
- [26] G. Tian, *Kähler–Einstein metrics with positive scalar curvature*, Invent. Math. 130 (1997), 1–37.
- [27] R. Vakil, *The rising sea: foundations of algebraic geometry*, Princeton University Press, Princeton, NJ, 2025.

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